

Concept 1 | Defining Angles and Basic Trigonometric Functions

English Student Edition • Focused formulas and guided practice

Formula opener

Radians

$$\pi \text{ rad} = 180^\circ$$

Arc length

$$S = r\theta$$

Trig

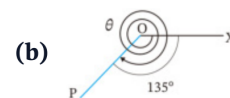
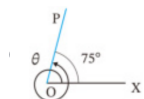
$$\sin \theta = \frac{y}{r}, \cos \theta = \frac{x}{r}$$

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q001 Written

Find the measures of θ in the two figures: (a) one full counter-clockwise revolution plus 75° ; (b) two full clockwise revolutions and a further 135° .



Classified Practice

Defining Angles and Basic Trigonometric Functions

Q002 **Written****Draw the terminal side for 300° and -450° .**

Q003 Written

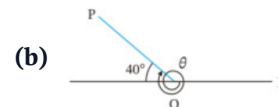
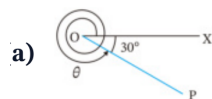
Express 420° and -210° as $\alpha + 360^\circ \times n$, where n is an integer and $0^\circ \leq \alpha < 360^\circ$.

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q004 Written

Find the measures of θ in the two figures: (a) a clockwise turn of 30° from the positive x -axis; (b) a terminal side 40° above the negative x -axis, measured counter-clockwise from the positive x -axis).



2 Draw the terminal side for the following angles.

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q005 **Written****Draw the terminal side for 500° and -120° .**

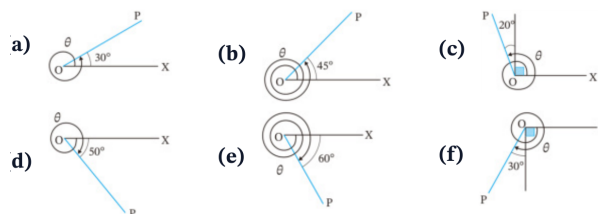
Q006 Written

Express 520° and -140° as $\alpha + 360^\circ \times n$.

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q007 Written

Find the six diagram measures θ shown in the given figure (a)–(f).

2 Draw the terminal side for the following angles.

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q008 Written

Draw the terminal sides for 470° , 780° , 225° , -380° , -990° , -200° .

Q009 Written

Express $610^\circ, 920^\circ, 840^\circ, -430^\circ, -960^\circ, -20^\circ$ **as** $\alpha + 360^\circ \times n$.

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q010 Written

Express 129720° and -2592100° as $\alpha + 360^\circ \times n$, with $0^\circ \leq \alpha < 360^\circ$.

Q011 MCQ

The principal angle coterminal with 765° is:**Choose the correct option.**A 45° B 135° C 225° D 315°

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q012 MCQ

The principal angle coterminal with -410° is:**Choose the correct option.**A -50° B 50° C 310° D 410°

Q013 MCQ**The terminal side of 540° lies on the:****Choose the correct option.**

- A positive x -axis
 - B negative x -axis
 - C positive y -axis
 - D negative y -axis
-
-
-

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q014 MCQ

A positive angle of 315° terminates in:**Choose the correct option.**

A QI

B QII

C QIII

D QIV

Q015 MCQ

Which is the general-angle form for -75° ?**Choose the correct option.**

- A $75^\circ + 360^\circ n$
- B $285^\circ + 360^\circ n$
- C $-285^\circ + 360^\circ n$
- D $360^\circ n - 75^\circ$

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q016 MCQ

The direction of a negative angle is:**Choose the correct option.**

- A counter-clockwise
- B clockwise
- C radial only
- D undefined

Q017 MCQ**The terminal side of 1080° is the:****Choose the correct option.**

- A positive x -axis
- B negative x -axis
- C positive y -axis
- D negative y -axis

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q018 MCQ

For $\alpha = 0^\circ$, the general-angle family is:**Choose the correct option.**A $360^\circ n$ B $0^\circ n$ C $180^\circ + 360^\circ n$ D $90^\circ + 360^\circ n$

Q019 MCQ**The principal angle for -990° is:****Choose the correct option.**A 90° B 180° C 270° D -270°

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q020 MCQ

A ray (P) is 25° above the negative x -axis. Its standard-position angle is:

Choose the correct option.

A 25°

B 155°

C 205°

D 335°

Q021 Written

(1) Convert 630° to radians. (2) Convert $-\frac{7\pi}{4}$ to degrees. (3) For a sector with $r = 3$ and $\theta = \frac{4\pi}{3}$, find arc length L and area S .

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q022 Written

Complete the standard-angle degree/radian table, then: convert $288^\circ, -400^\circ$ to radians; convert $\frac{11\pi}{6}, -\frac{3\pi}{5}$ to degrees; and find L, S for $r = 6, \theta = \frac{3\pi}{4}$.

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q023 Written

Convert $108^\circ, 36^\circ, -240^\circ, -225^\circ, 420^\circ, 400^\circ, -570^\circ, -2700^\circ$ to radians.

Q024 Written

Convert $\frac{7\pi}{6}, \frac{4\pi}{5}, \frac{2\pi}{3}, \frac{4\pi}{3}, -\frac{5\pi}{4}, -\frac{\pi}{2}, -\frac{11\pi}{2}, -\frac{7\pi}{18}$ **to degrees.**

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q025 Written

Find L and S for: (a) $r = 6, \theta = \frac{7\pi}{6}$; (b) $r = 4, \theta = \frac{2\pi}{5}$; (c) $r = 6, \theta = \frac{5\pi}{4}$.

Q026 Written

Draw terminal sides for $\frac{\pi}{4}, \frac{\pi}{3}, \frac{5\pi}{6}, \frac{3\pi}{4}, \frac{4\pi}{3}, \frac{7\pi}{6}, -\frac{\pi}{3}, -\frac{\pi}{4}$.

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q027 Written

A sector of radius 9 cm and central angle $4\pi/3$ is joined to form a cone. Determine the exact vertical height and total volume.

Q028 MCQ

Convert 150° to radians:**Choose the correct option.**

- A $\frac{5\pi}{6}$
B $\frac{3\pi}{5}$
C $\frac{5\pi}{3}$
D $\frac{3\pi}{6}$

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q029 MCQ

Convert $-\frac{5\pi}{6}$ to degrees:**Choose the correct option.**A -120° B -150° C 150° D -300°

Q030 MCQ

For $r = 5$, $\theta = \frac{2\pi}{3}$, the arc length is:

Choose the correct option.

- A $\frac{5\pi}{3}$
- B $\frac{10\pi}{3}$
- C $\frac{25\pi}{3}$
- D 10π

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q031 MCQ

For $r = 4$, $\theta = \frac{\pi}{2}$, the sector area is:

Choose the correct option.

- A π
- B 2π
- C 4π
- D 8π

Q032 MCQ

The radian measure of a full turn is:**Choose the correct option.**

- A π
 - B 2π
 - C 180
 - D 360
-
-
-

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q033 MCQ

The degree measure of $\frac{7\pi}{4}$ is:

Choose the correct option.

- A 210°
- B 270°
- C 315°
- D 360°

Q034 MCQ

The principal angle for $-\frac{11\pi}{6}$ is:

Choose the correct option.

- A $\frac{\pi}{6}$
- B $-\frac{\pi}{6}$
- C $\frac{5\pi}{6}$
- D $\frac{11\pi}{6}$

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q035 MCQ

For $r = 3, \theta = \pi$, the sector area is:**Choose the correct option.**

A $\frac{3\pi}{2}$

B 3π

C 9π

D $\frac{9\pi}{2}$

Q036 MCQ**The angle $\frac{5\pi}{3}$ lies in:****Choose the correct option.**

A QI

B QII

C QIII

D QIV

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q037 Written

Convert 252° to radians.

Q038 Written

A sector has $r = 8$ and $\theta = \frac{3\pi}{4}$. Find L and S .

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q039 Written

If $\theta = \frac{4\pi}{3}$, find $\sin \theta$, $\cos \theta$, $\tan \theta$.

Q040 Written

Find the exact values (write undefined where appropriate): $\sin \frac{7\pi}{4}$, $\cos \frac{\pi}{3}$, $\tan \frac{\pi}{6}$, $\sin(-\frac{2\pi}{3})$, $\cos(-\pi)$, $\tan(-\frac{\pi}{2})$.

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q041 Written

Find the exact values of the eighteen trigonometric functions listed in the given exercise (a)–(r), including negative and undefined cases.

Find the values of the following trigonometric functions. If the value is undefined, write “undefined”.

(a) $\sin \frac{\pi}{3}$

(b) $\sin \frac{7}{6} \pi$

(c) $\sin \pi$

(d) $\cos \frac{5}{4} \pi$

(e) $\cos \frac{5}{3} \pi$

(f) $\cos \frac{7}{6} \pi$

(g) $\tan \frac{2}{3} \pi$

(h) $\tan \frac{5}{4} \pi$

(i) $\tan \pi$

(j) $\sin \left(-\frac{\pi}{6} \right)$

(k) $\sin \left(-\frac{5}{4} \pi \right)$

(l) $\sin \left(-\frac{5}{3} \pi \right)$

(m) $\cos \left(-\frac{\pi}{4} \right)$

(n) $\cos \left(-\frac{5}{6} \pi \right)$

(o) $\cos \left(-\frac{4}{3} \pi \right)$

(p) $\tan \left(-\frac{4}{3} \pi \right)$

(q) $\tan \left(-\frac{11}{6} \pi \right)$

(r) $\tan \left(-\frac{3}{2} \pi \right)$

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q042 Written

The panel height is $h(t) = 3 \sin\left(\frac{2\pi}{3}t - \frac{\pi}{4}\right) + 1.5$. Find the exact height at $t = 0$.

Q043 MCQ

 $\sin \frac{\pi}{6}$ equals:

Choose the correct option.

A 0

B $\frac{1}{2}$ C $\frac{\sqrt{3}}{2}$

D 1

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q044 MCQ

 $\cos \frac{2\pi}{3}$ equals:

Choose the correct option.

A -1

B $-\frac{1}{2}$ C $\frac{1}{2}$ D $\frac{\sqrt{3}}{2}$

Q045 MCQ $\tan \frac{3\pi}{4}$ equals:

Choose the correct option.

A -1

B 0

C 1

D undefined

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q046 MCQ

 $\sin\left(-\frac{\pi}{2}\right)$ equals:

Choose the correct option.

- A -1
- B 0
- C 1
- D undefined

Q047 MCQ $\tan \frac{\pi}{2}$ is:**Choose the correct option.**

A 0

B 1

C -1

D undefined

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q048 MCQ

 $\cos \frac{5\pi}{3}$ equals:

Choose the correct option.

A $-\frac{1}{2}$

B $\frac{1}{2}$

C $-\frac{\sqrt{3}}{2}$

D $\frac{\sqrt{3}}{2}$

Q049 MCQ**If $P(x, y)$ lies on the unit circle, then $\sin \theta$ is:****Choose the correct option.****A** x **B** y **C** x/y **D** $1/x$

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q050 MCQ

 $\tan\left(-\frac{3\pi}{4}\right)$ equals:

Choose the correct option.

A -1

B 0

C 1

D undefined

Q051 MCQ $\sin \frac{11\pi}{6}$ equals:

Choose the correct option.

A $-\frac{1}{2}$

B $\frac{1}{2}$

C $-\frac{\sqrt{3}}{2}$

D $\frac{\sqrt{3}}{2}$

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q052 Written

Find $\sin \frac{5\pi}{6}$, $\cos \frac{4\pi}{3}$, $\tan \frac{7\pi}{4}$.

Q053 Written

Find $\sin(-\frac{5\pi}{6})$, $\cos(-\frac{7\pi}{6})$, $\tan(-\frac{\pi}{3})$.

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q054 **Written**

For $P = \left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$ on the unit circle, find the corresponding $\sin \theta$, $\cos \theta$, $\tan \theta$.

Q055 Written

Find the exact value of $3 \sin \frac{3\pi}{2} - 2 \cos \pi$.

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q056 Written

State the two standard angles where $\tan \theta$ is undefined in $[0, 2\pi)$.

Q057 Written

Solve for θ , $0 \leq \theta < 2\pi$: (a) $\sin \theta = -\frac{1}{2}$; (b) $\cos \theta = \frac{1}{\sqrt{2}}$; (c) $\tan \theta = -\sqrt{3}$.

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q058 Written

Solve the six trigonometric equations in the given Try set for $0 \leq \theta < 2\pi$: $\sin \theta = 1/\sqrt{2}$, $\cos \theta = -\sqrt{3}/2$, $\tan \theta = 1/\sqrt{3}$, $\sin \theta = 0$, $\cos \theta = 1/2$, $\tan \theta = -1$.

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q059 **Written**

Solve the sixteen equations in the given exercise set (a)–(p) for $0 \leq \theta < 2\pi$.

Q060 Written**Solve** $\tan^2 \theta + (1 - \sqrt{3}) \tan \theta = \sqrt{3}$ for $0 \leq \theta < 360^\circ$.

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q061 MCQ

For $0 \leq \theta < 2\pi$, $\sin \theta = \frac{1}{2}$ gives:

Choose the correct option.

A $\frac{\pi}{6}, \frac{5\pi}{6}$

B $\frac{\pi}{3}, \frac{2\pi}{3}$

C $\frac{\pi}{6}, \frac{7\pi}{6}$

D $\frac{5\pi}{6}, \frac{11\pi}{6}$

Q062 MCQ

For $0 \leq \theta < 2\pi$, $\cos \theta = -\frac{1}{2}$ gives:

Choose the correct option.

A $\frac{\pi}{3}, \frac{5\pi}{3}$

B $\frac{2\pi}{3}, \frac{4\pi}{3}$

C $\frac{\pi}{6}, \frac{5\pi}{6}$

D $\frac{4\pi}{3}, \frac{5\pi}{3}$

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q063 MCQ

For $0 \leq \theta < 2\pi$, $\tan \theta = 1$ gives:**Choose the correct option.**

- A $\frac{\pi}{4}, \frac{5\pi}{4}$
B $\frac{3\pi}{4}, \frac{7\pi}{4}$
C $\frac{\pi}{4}, \frac{7\pi}{4}$
D $\frac{3\pi}{4}, \frac{5\pi}{4}$

Q064 MCQ

The solutions of $\sin \theta = 0$ in $[0, 2\pi)$ are:**Choose the correct option.**A $0, \pi$ B $\frac{\pi}{2}, \frac{3\pi}{2}$ C $0, 2\pi$ D π

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q065 MCQ

If $\cos \theta = 1$, then in $[0, 2\pi)$, θ is:

Choose the correct option.

A 0

B $\frac{\pi}{2}$ C π D $\frac{3\pi}{2}$

Q066 MCQ**For $0 \leq \theta < 2\pi$, $\tan \theta = -\sqrt{3}$ gives:****Choose the correct option.**

- A $\frac{\pi}{3}, \frac{4\pi}{3}$
B $\frac{2\pi}{3}, \frac{5\pi}{3}$
C $\frac{\pi}{6}, \frac{7\pi}{6}$
D $\frac{5\pi}{6}, \frac{11\pi}{6}$
-
-
-

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q067 MCQ

For $0 \leq \theta < 2\pi$, $\sin \theta = -1$ gives:

Choose the correct option.

- A $\frac{\pi}{2}$
- B $\frac{3\pi}{2}$
- C $\frac{2}{\pi}$
- D 0

Q068 MCQ**For $0 \leq \theta < 2\pi$, $\cos \theta = 0$ gives:****Choose the correct option.**

- A $0, \pi$
- B $\frac{\pi}{2}, \frac{3\pi}{2}$
- C π
- D $\frac{\pi}{4}, \frac{5\pi}{4}$

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q069 MCQ

The equation $\tan \theta = 0$ has solutions in $[0, 2\pi)$:

Choose the correct option.

A $0, \pi$ B $\frac{\pi}{2}, \frac{3\pi}{2}$ C $\frac{\pi}{4}, \frac{5\pi}{4}$ D $\pi, 2\pi$

Q070 Written**Solve** $\cos \theta = \frac{\sqrt{3}}{2}$ **for** $0 \leq \theta < 2\pi$.

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q071 Written

Solve $\sin \theta = -\frac{\sqrt{3}}{2}$ for $0 \leq \theta < 2\pi$.

Q072 Written**Solve** $\tan \theta = \frac{1}{\sqrt{3}}$ for $0 \leq \theta < 2\pi$.

Classified Practice

Defining Angles and Basic Trigonometric Functions

Q073 Written

Solve $2 \cos \theta - 1 = 0$ for $0 \leq \theta < 2\pi$.

Q074 Written**Solve** $2 \sin \theta + \sqrt{2} = 0$ for $0 \leq \theta < 2\pi$.

Classified Practice

Defining Angles and Basic Trigonometric Functions

Quiz MCQ 1 [Concept Quiz · MCQ](#)

The radian measure of 240° is:

Choose the correct option.

A $\frac{2\pi}{3}$

B $\frac{4\pi}{3}$

C $\frac{3\pi}{4}$

D $\frac{5\pi}{3}$

Quiz MCQ 2 Concept Quiz · MCQ

If $\sin \theta = -\frac{\sqrt{3}}{2}$, $0 \leq \theta < 2\pi$, then θ is:

Choose the correct option.

- A $\frac{\pi}{3}, \frac{2\pi}{3}$
B $\frac{4\pi}{3}, \frac{5\pi}{3}$
C $\frac{2\pi}{3}, \frac{4\pi}{3}$
D $\frac{5\pi}{6}, \frac{11\pi}{6}$

Classified Practice

Defining Angles and Basic Trigonometric Functions

Quiz MCQ 3 **Concept Quiz · MCQ**

For a sector with $r = 6$ and $\theta = \frac{\pi}{3}$, the area is:

Choose the correct option.

A 3π

B 6π

C 12π

D 18π

Quiz MCQ 4 Concept Quiz · MCQ

The general-angle family for -300° is:

Choose the correct option.

- A $60^\circ + 360^\circ n$
- B $300^\circ + 360^\circ n$
- C $-60^\circ + 360^\circ n$
- D $120^\circ + 360^\circ n$

Classified Practice

Defining Angles and Basic Trigonometric Functions

Quiz WRITTEN 5 Concept Quiz · Written**Solve** $\cos \theta = -\frac{\sqrt{2}}{2}$ **for** $0 \leq \theta < 2\pi$.

Quiz WRITTEN 6 Concept Quiz · Written

Convert $-\frac{13\pi}{6}$ to degrees and state its principal angle.

Classified Practice

Defining Angles and Basic Trigonometric Functions

Quiz WRITTEN 7 Concept Quiz · Written

Find L and S for a sector with $r = 10$ and $\theta = \frac{2\pi}{5}$.

Quiz WRITTEN 8 Concept Quiz · Written

Evaluate $3 \sin \frac{7\pi}{6} - 2 \cos \frac{4\pi}{3}$.

Use the question number shown on each page.

Concept 2 | Trigonometric Identities and Expression Evaluation

English Student Edition • Focused formulas and guided practice

Formula opener

Identity

$$\sin^2 \theta + \cos^2 \theta = 1$$

Tangent

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

Secant

$$1 + \tan^2 \theta = \sec^2 \theta$$

Classified Practice

Trigonometric Identities and Expression Evaluation

Q001 Written

Given (a) $\sin \theta = -\frac{1}{5}$ in QIII and (b) $\tan \theta = -\sqrt{6}$ with $\frac{3\pi}{2} < \theta < 2\pi$, find the other two trigonometric values in each case.

Classified Practice

Trigonometric Identities and Expression Evaluation

Q002 Written

Given one value and the quadrant, find the other two: (a) $\cos \theta = -1/\sqrt{3}$, QIII; (b) $\tan \theta = -2$, QIV; (c) $\sin \theta = 3/5$, QII; (d) $\tan \theta = \sqrt{3}/2$, $\pi < \theta < 3\pi/2$.

Classified Practice

Trigonometric Identities and Expression Evaluation

Q003 Written

For each of the twelve numbered cases in the given Exercise, one of $\sin \theta$, $\cos \theta$, $\tan \theta$ and a quadrant/interval is given. Find the other two values.

Q004 Written

Using trigonometric identities, find the other two values given $\tan \theta + \cot \theta = 12$ and $\pi < \theta < 2\pi$.

Classified Practice

Trigonometric Identities and Expression Evaluation

Q005 **Written**

Given $\sin \theta = \frac{5}{13}$ **in QII, find** $\cos \theta$ **and** $\tan \theta$.

Q006 Written

Given $\cos \theta = -\frac{3}{5}$ **in QIII, find** $\sin \theta$ **and** $\tan \theta$.

Classified Practice

Trigonometric Identities and Expression Evaluation

Q007 **Written**

Given $\tan \theta = \frac{3}{4}$ **in QI, find** $\sin \theta$ **and** $\cos \theta$.

Q008 Written

Given $\tan \theta = -\frac{5}{12}$ in QIV, find $\sin \theta$ and $\cos \theta$.

Classified Practice

Trigonometric Identities and Expression Evaluation

Q009 Written

Given $\sin \theta = -\frac{\sqrt{2}}{2}$ in QIII, find $\cos \theta$ and $\tan \theta$.

Q010 MCQ

If $\sin \theta = -\frac{3}{5}$ in QIII, $\cos \theta$ is:

Choose the correct option.

A $\frac{4}{5}$

B $-\frac{4}{5}$

C $\frac{3}{5}$

D $-\frac{3}{5}$

Classified Practice

Trigonometric Identities and Expression Evaluation

Q011 MCQ

If $\tan \theta = 2$ in QI, $\sin \theta$ is:

Choose the correct option.

A $\frac{2\sqrt{5}}{5}$

B $\frac{\sqrt{5}}{5}$

C $-\frac{2\sqrt{5}}{5}$

D $\frac{2}{5}$

Q012 MCQ

If $\cos \theta = \frac{1}{\sqrt{2}}$ in QIV, $\tan \theta$ is:

Choose the correct option.

A 1

B -1

C $\sqrt{2}$

D $-\sqrt{2}$

Classified Practice

Trigonometric Identities and Expression Evaluation

Q013 MCQ

For $\tan \theta = -\sqrt{3}$ in QII, $\sin \theta$ is:

Choose the correct option.

- A $\frac{1}{2}$
- B $-\frac{1}{2}$
- C $\frac{\sqrt{3}}{2}$
- D $-\frac{\sqrt{3}}{2}$

Q014 MCQ

If $\sin \theta = \frac{7}{25}$ in QI, $\tan \theta$ is:

Choose the correct option.

- A $\frac{7}{24}$
- B $\frac{24}{7}$
- C $-\frac{7}{24}$
- D $\frac{25}{7}$

Classified Practice

Trigonometric Identities and Expression Evaluation

Q015 Written

Given $\sin \theta + \cos \theta = \frac{1}{2}$, **with** θ **in** QII, **evaluate** (a) $\sin \theta \cos \theta$, (b) $\sin^3 \theta + \cos^3 \theta$, (c) $\sin \theta - \cos \theta$.

Q016 Written

Given $\sin \theta + \cos \theta = \frac{1}{3}$, **with** θ **in** QII, **evaluate** (a) $\sin \theta \cos \theta$, (b) $\sin^3 \theta + \cos^3 \theta$, (c) $\sin \theta - \cos \theta$.

Classified Practice

Trigonometric Identities and Expression Evaluation

Q017 Written

Solve the three numbered Exercise problems: (1) given $\sin \theta - \cos \theta = \sqrt{3}/2$ in QI; (2) given $\sin \theta + \cos \theta = \sqrt{3}/2$ in QIV; (3) given $\sin \theta - \cos \theta = 1/2$ in QIII. Evaluate the three expressions requested in each case.

Classified Practice

Trigonometric Identities and Expression Evaluation

Q018 Written

Given $\sin \theta - \cos \theta = 0.5$ **and** θ **in** QIII, **evaluate the exact value of** $\sin^6 \theta - \cos^6 \theta$.

Q019 Written

Given $\sin \theta + \cos \theta = \frac{2}{3}$ **in QII, find** $\sin \theta \cos \theta$ **and** $\sin \theta - \cos \theta$.

Classified Practice

Trigonometric Identities and Expression Evaluation

Q020 Written

Given $\sin \theta - \cos \theta = \frac{\sqrt{5}}{2}$ in QI, find $\sin \theta \cos \theta$ and $\sin^3 \theta - \cos^3 \theta$.

Q021 Written

Given $\sin \theta + \cos \theta = -\frac{1}{2}$ **in QIII, find** $\sin^3 \theta + \cos^3 \theta$.

Classified Practice

Trigonometric Identities and Expression Evaluation

Q022 Written

Given $\sin \theta - \cos \theta = \frac{1}{3}$ **in QIV, find** $\sin \theta + \cos \theta$.

Q023 Written**Given** $\sin \theta - \cos \theta = 1$ **in QIII, find** $\sin^3 \theta - \cos^3 \theta$.

Classified Practice

Trigonometric Identities and Expression Evaluation

Q024 MCQ

If $s + c = \frac{1}{2}$, then sc equals:

Choose the correct option.

A $-\frac{3}{8}$

B $\frac{3}{8}$

C $-\frac{1}{4}$

D $\frac{1}{4}$

Q025 MCQ

If $s + c = \frac{1}{2}$, then $s^3 + c^3$ equals:

Choose the correct option.

- A $\frac{5}{16}$
B $\frac{11}{16}$
C $-\frac{11}{16}$
D $\frac{3}{16}$

Classified Practice

Trigonometric Identities and Expression Evaluation

Q026 MCQ

If $s - c = \frac{1}{2}$ in QIII, then $s + c$ equals:

Choose the correct option.

A $\frac{\sqrt{7}}{2}$

B $-\frac{\sqrt{7}}{2}$

C $\frac{1}{2}$

D $-\frac{1}{2}$

Q027 MCQ

If $s - c = \frac{\sqrt{3}}{2}$ in QI, then sc equals:

Choose the correct option.

A $\frac{1}{8}$

B $-\frac{1}{8}$

C $\frac{3}{8}$

D $-\frac{3}{8}$

Classified Practice

Trigonometric Identities and Expression Evaluation

Q028 MCQ

If $s - c = 0.5$ in QIII, then $s^6 - c^6$ equals:

Choose the correct option.

A $-\frac{55\sqrt{7}}{256}$

B $\frac{55\sqrt{7}}{256}$

C $-\frac{11\sqrt{7}}{16}$

D $\frac{11}{16}$

Q029 Written

Prove $\frac{\sin \theta}{1+\cos \theta} + \frac{1}{\tan \theta} = \frac{1}{\sin \theta}.$

Classified Practice

Trigonometric Identities and Expression Evaluation

Q030 Written

Prove (a) $\frac{1-\tan \theta}{1+\tan \theta} = \frac{\cos \theta - \sin \theta}{\cos \theta + \sin \theta}$; (b) $\frac{\cos \theta}{1-\sin \theta} - \tan \theta = \frac{1}{\cos \theta}$.

Classified Practice

Trigonometric Identities and Expression Evaluation

Q031 Written

Prove the six identities in the given Exercise: (a) $\tan^2 \theta - \sin^2 \theta = \tan^2 \theta \sin^2 \theta$; (b) $\tan \theta + 1/\tan \theta = 1/(\sin \theta \cos \theta)$; (c) $\cos \theta/(1 - \sin \theta) - 1/\cos \theta = \tan \theta$; (d) $\tan \theta/\sin \theta - \sin \theta/\tan \theta = \sin \theta \tan \theta$; (e) $(1 + \sin \theta)/\cos \theta + \cos \theta/(1 + \sin \theta) = 2/\cos \theta$; (f) $(\tan^2 \theta - \sin^2 \theta)/\sin^2 \theta = \tan^2 \theta$.

Classified Practice

Trigonometric Identities and Expression Evaluation

Q032 Written

Prove that, for all defined values of x , $\frac{1+\sin x-\cos x}{1+\sin x+\cos x} + \frac{1+\sin x+\cos x}{1+\sin x-\cos x} = 2 \csc x$.

Q033 Written

Prove $\frac{1+\sin \theta}{\cos \theta} = \frac{\cos \theta}{1-\sin \theta}$.

Classified Practice

Trigonometric Identities and Expression Evaluation

Q034 Written

Prove $\frac{1}{1+\cos\theta} + \frac{1}{1-\cos\theta} = 2\sec^2\theta$.

Q035 Written

Prove $\frac{\tan \theta}{1 + \tan \theta} = \frac{\sin \theta}{\sin \theta + \cos \theta}$.

Classified Practice

Trigonometric Identities and Expression Evaluation

Q036 Written

Prove $\frac{\sin \theta}{1 - \cos \theta} = \frac{1 + \cos \theta}{\sin \theta}.$

Q037 Written

Prove $\frac{\cos \theta}{1 + \sin \theta} = \frac{1 - \sin \theta}{\cos \theta}$.

Classified Practice

Trigonometric Identities and Expression Evaluation

Q038 MCQ

The first move in proving a trig identity is usually to:**Choose the correct option.**

- A Change both sides at once
- B Simplify one side
- C Substitute random values
- D Take a square root

Q039 MCQ**The identity used to replace $1 - \sin^2 \theta$ is:****Choose the correct option.**A $\cos^2 \theta$ B $\tan^2 \theta$ C $\sec^2 \theta$ D $\sin^2 \theta$

Classified Practice

Trigonometric Identities and Expression Evaluation

Q040 MCQ

$\tan \theta$ should be rewritten as:

Choose the correct option.

A $\sin \theta \cos \theta$

B $\sin \theta / \cos \theta$

C $\cos \theta / \sin \theta$

D $1 / \sin \theta$

Q041 MCQ**To add two trig fractions, first find a:****Choose the correct option.**

- A common numerator
- B common denominator
- C new quadrant
- D reference angle

Classified Practice

Trigonometric Identities and Expression Evaluation

Q042 MCQ

The result of a valid identity proof is:**Choose the correct option.**

- A LHS=0
- B LHS=RHS
- C RHS=1
- D undefined

Q043 MCQ

If $\cos \theta = -\frac{12}{13}$ in QII, $\sin \theta$ is:

Choose the correct option.

A $\frac{5}{13}$

B $-\frac{5}{13}$

C $\frac{12}{13}$

D $-\frac{12}{5}$

Classified Practice

Trigonometric Identities and Expression Evaluation

Q044 MCQ

If $\tan \theta = -\frac{1}{2}$ in QII, $\cos \theta$ is:

Choose the correct option.

A $\frac{2\sqrt{5}}{5}$

B $-\frac{2\sqrt{5}}{5}$

C $\frac{\sqrt{5}}{5}$

D $-\frac{\sqrt{5}}{5}$

Q045 MCQ

If $\sin \theta = \frac{8}{17}$ in QIV, $\tan \theta$ is:

Choose the correct option.

A $\frac{8}{15}$

B $-\frac{8}{15}$

C $\frac{15}{8}$

D $-\frac{15}{8}$

Classified Practice

Trigonometric Identities and Expression Evaluation

Q046 MCQ

If $\tan \theta = \sqrt{3}$ and θ is in QIII, $\sin \theta$ is:

Choose the correct option.

- A $\frac{1}{2}$
- B $-\frac{1}{2}$
- C $\frac{\sqrt{3}}{2}$
- D $-\frac{\sqrt{3}}{2}$

Q047 MCQ**If $s + c = 1$, then sc equals:****Choose the correct option.****A** 0**B** $\frac{1}{2}$ **C** $-\frac{1}{2}$ **D** 1

Classified Practice

Trigonometric Identities and Expression Evaluation

Q048 MCQ

If $s - c = \sqrt{2}$, then sc equals:

Choose the correct option.

A $-\frac{1}{2}$

B $\frac{1}{2}$

C -1

D 1

Q049 MCQ

If $s + c = \frac{2}{3}$, then $s^3 + c^3$ equals:

Choose the correct option.

- A $\frac{19}{27}$
B $\frac{5}{27}$
C $-\frac{19}{27}$
D $\frac{11}{27}$

Classified Practice

Trigonometric Identities and Expression Evaluation

Q050 MCQ

If $s - c = \frac{1}{2}$ in QIII, then $s^3 - c^3$ equals:

Choose the correct option.

A $\frac{11}{16}$

B $-\frac{11}{16}$

C $\frac{7}{16}$

D $-\frac{7}{16}$

Q051 MCQ

Which identity converts $\frac{1}{\tan \theta}$ into sine and cosine?

Choose the correct option.

- A $\frac{\sin \theta}{\cos \theta}$
- B $\frac{\cos \theta}{\sin \theta}$
- C $\sin \theta \cos \theta$
- D $\sec \theta$

Classified Practice

Trigonometric Identities and Expression Evaluation

Q052 MCQ

The identity $1 - \cos^2 \theta$ simplifies to:**Choose the correct option.**A $\sin^2 \theta$ B $\tan^2 \theta$ C $\cos^2 \theta$

D 1

Q053 MCQ**When proving an identity, which side should normally be transformed?****Choose the correct option.**

- A Both sides simultaneously
- B One side only
- C Neither side
- D Only the denominator

Classified Practice

Trigonometric Identities and Expression Evaluation

Q054 MCQ

After combining trig fractions, the next step is usually to use:**Choose the correct option.**

- A A random angle
- B The Pythagorean identity
- C A logarithm
- D A derivative

Quiz MCQ 1 Concept Quiz · MCQ

If $\sin \theta + \cos \theta = \frac{1}{2}$, then $\sin \theta \cos \theta$ is:

Choose the correct option.

A $-\frac{3}{8}$

B $\frac{3}{8}$

C $-\frac{1}{4}$

D $\frac{1}{4}$

Classified Practice

Trigonometric Identities and Expression Evaluation

Quiz MCQ 2 **Concept Quiz · MCQ**

If $\tan \theta = \frac{3}{4}$ in QI, then $\cos \theta$ is:

Choose the correct option.

- A $\frac{3}{5}$
- B $\frac{4}{5}$
- C $-\frac{4}{5}$
- D $\frac{3}{4}$

Quiz MCQ 3 Concept Quiz · MCQ

Which identity is always valid where defined?

Choose the correct option.

A $\tan \theta = \frac{\cos \theta}{\sin \theta}$

B $1 + \tan^2 \theta = \frac{1}{\cos^2 \theta}$

C $\sin \theta + \cos \theta = 1$

D $\sin \theta \cos \theta = 1$

Classified Practice

Trigonometric Identities and Expression Evaluation

Quiz MCQ 4 Concept Quiz · MCQ

If $s - c = \frac{1}{2}$ in QIII, then $s + c$ is:

Choose the correct option.

A $\frac{\sqrt{7}}{2}$

B $-\frac{\sqrt{7}}{2}$

C $\frac{1}{2}$

D $-\frac{1}{2}$

Quiz WRITTEN 5 Concept Quiz · Written

Given $\cos \theta = -\frac{5}{13}$ **in QIII, find** $\sin \theta$ **and** $\tan \theta$.

Classified Practice

Trigonometric Identities and Expression Evaluation

Quiz WRITTEN 6 Concept Quiz · Written**Given** $s + c = \frac{2}{3}$, **find** sc **and** $s^3 + c^3$.

Quiz WRITTEN 7 Concept Quiz · Written

Prove $\frac{1-\sin \theta}{\cos \theta} = \frac{\cos \theta}{1+\sin \theta}$.

Classified Practice

Trigonometric Identities and Expression Evaluation

Quiz WRITTEN 8 Concept Quiz · Written

Given $\tan \theta = -2$ **in QIV, find** $\sin \theta$ **and** $\cos \theta$, **then verify** $1 + \tan^2 \theta = 1/\cos^2 \theta$.

Answer key

Use the question number shown on each page.

Q001 (a) $\cos \theta = -\frac{2\sqrt{6}}{5}$, $\tan \theta = \frac{\sqrt{6}}{12}$; (b) $\cos \theta = \frac{\sqrt{7}}{7}$, $\sin \theta = -\frac{\sqrt{42}}{7}$ **Q015** (a) $-\frac{3}{8}$ (b) $\frac{11}{16}$ (c) $\frac{\sqrt{7}}{2}$

Q002 (a) $\sin \theta = -\frac{\sqrt{6}}{3}$, $\tan \theta = \sqrt{2}$; (b) $\cos \theta = \frac{\sqrt{5}}{5}$, $\sin \theta = -\frac{2\sqrt{5}}{5}$; (c) $\cos \theta = -\frac{4}{5}$, $\tan \theta = -\frac{3}{4}$; (d) $\sin \theta = -\frac{\sqrt{21}}{7}$, $\cos \theta = -\frac{2\sqrt{7}}{7}$

Q003 $\sin \theta = \frac{\sqrt{2}}{2}$, $\cos \theta = \frac{\sqrt{2}}{2}$; (b) $\sin \theta = \frac{\sqrt{2}}{2}$, $\cos \theta = \frac{\sqrt{2}}{2}$; (c) $\sin \theta = \frac{\sqrt{2}}{2}$, $\cos \theta = \frac{\sqrt{2}}{2}$; (d) $\sin \theta = \frac{\sqrt{2}}{2}$, $\cos \theta = \frac{\sqrt{2}}{2}$ **Q016** (a) $-\frac{4}{9}$ (b) $\frac{37}{27}$ (c) $\frac{\sqrt{17}}{3}$

Q004 $\sin \theta = -\frac{1}{2}$, $\cos \theta = -\frac{\sqrt{3}}{2}$ or the swapped pair **Q017** (1) $\sin \theta \cos \theta = \frac{1}{8}$, $\sin^2 \theta - \cos^2 \theta = \frac{7\sqrt{3}}{16}$, $\sin \theta + \cos \theta = \frac{\sqrt{5}}{2}$; (2) $-\frac{1}{8}$, $\frac{9\sqrt{3}}{16}$, $-\frac{\sqrt{5}}{2}$; (3) $\frac{3}{8}$, $\frac{11}{16}$ **Q031** LHS = RHS for (a)+(f)

Q005 $-\frac{12}{13}$, $-\frac{5}{12}$ **Q018** $-\frac{55\sqrt{7}}{256}$ **Q032** LHS = $2 \csc x$

Q006 $-\frac{4}{5}$, $\frac{4}{3}$ **Q019** $-\frac{5}{18}$, $\frac{\sqrt{14}}{3}$ **Q033** Identity proved

Q007 $\frac{3}{5}$, $\frac{4}{5}$ **Q020** $-\frac{1}{8}$, $\frac{7\sqrt{5}}{16}$ **Q034** Identity proved

Q008 $-\frac{5}{13}$, $\frac{12}{13}$ **Q021** $-\frac{11}{16}$ **Q035** Identity proved

Q009 $-\frac{\sqrt{2}}{2}$, 1 **Q022** $\frac{\sqrt{11}}{3}$ **Q036** Identity proved

Q010 B **Q023** $\frac{3}{2}$ **Q037** Identity proved

Q011 A **Q024** A **Q038** B

Q012 B **Q025** B **Q039** A

Q013 C **Q026** B **Q040** B

Q014 A **Q027** A **Q041** B

Q028 A

Q029 LHS = RHS

Q030 Both identities verified

Q031 LHS = RHS for (a)+(f)

Q032 LHS = $2 \csc x$

Q033 Identity proved

Q034 Identity proved

Q035 Identity proved

Q036 Identity proved

Q037 Identity proved

Q038 B

Q039 A

Q040 B

Q041 B

Q042 B

Q043 A

Q044 A

Q045 B

Q046 D

Q047 A

Q048 A

Q049 A

Q050 A

Q051 B

Q052 A

Q053 B

Q054 B

Quiz MCQ 1 A

Quiz MCQ 2 B

Quiz MCQ 3 B

Quiz MCQ 4 B

Quiz WRITTEN 5 $-\frac{12}{13}$, $\frac{12}{5}$

Quiz WRITTEN 6 $-\frac{5}{18}$, $\frac{19}{27}$

Quiz WRITTEN 7 Identity proved

Quiz WRITTEN 8 $\sin \theta = -\frac{2\sqrt{5}}{5}$, $\cos \theta = \frac{\sqrt{5}}{5}$; $6 = 6$

Concept 3 | Properties and Graphs of Trigonometric Functions

English Student Edition • Focused formulas and guided practice

Formula opener

Amplitude

$$y = a \sin(bx + c) + d \Rightarrow |a|$$

Period

$$T = \frac{2\pi}{|b|}$$

Range

$$d - |a| \leq y \leq d + |a|$$

Classified Practice

Properties and Graphs of Trigonometric Functions

Q001 Written

(1) Express $\sin \frac{9\pi}{5}$ as a trigonometric function of an acute angle. (2) Simplify $\sin \theta + \sin(\theta + \frac{\pi}{2}) + \sin(\theta + \pi) + \sin(\theta + \frac{3\pi}{2})$.

Classified Practice

Properties and Graphs of Trigonometric Functions

Q002 Written

(1) Express $\cos(-\frac{\pi}{6})$ as a trigonometric function of an acute angle. (2) Express $\sin \frac{8\pi}{7}$ as a trigonometric function of an acute angle. (3) Simplify $\cos(\frac{\pi}{2} - \theta) \sin(\pi - \theta) - \sin(\frac{3\pi}{2} + \theta) \cos(\pi - \theta)$.

Classified Practice

Properties and Graphs of Trigonometric Functions

Q003 Written

Solve all three given demands: (1) express $\cos \frac{17\pi}{5}$, $\cos \frac{5\pi}{7}$, $\sin(-\frac{7\pi}{8})$, $\sin(-\frac{19\pi}{8})$ using acute angles; (2) find $\tan(-\frac{\pi}{6})$ and $\tan(-\frac{\pi}{4})$; (3) simplify (a) $\sin \theta + \cos(\pi + \theta) - \cos(\pi - \theta) + \sin(\pi + \theta)$, (b) $\sin(\theta + \frac{\pi}{2}) \cos(\theta + \pi) + \sin(-\theta) \cos(\frac{\pi}{2} - \theta)$.

Classified Practice

Properties and Graphs of Trigonometric Functions

Q004 **Written**

Simplify $\sin\left(\frac{3\pi}{2} - \theta\right) \sec(\pi + \theta) - \tan\left(\theta - \frac{\pi}{2}\right) \cot\left(\frac{5\pi}{2} + \theta\right).$

Q005 Written

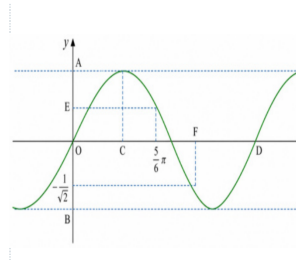
Draw the parent graphs $y = \sin \theta$, $y = \cos \theta$, and $y = \tan \theta$, and state each period and range.

Classified Practice

Properties and Graphs of Trigonometric Functions

Q006 Written

For the given graph of $y = \sin \theta$, determine the labelled scales A to F .

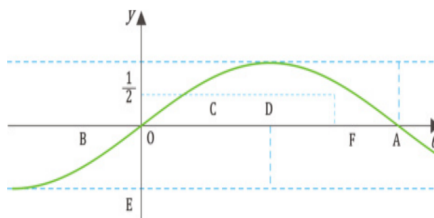


Classified Practice

Properties and Graphs of Trigonometric Functions

Q007 Written

(1) Draw the parent graphs $y = \sin \theta$, $y = \cos \theta$, $y = \tan \theta$, and state their periods. (2) For the given graph of $y = \cos \theta$, determine the labelled scales A to F .

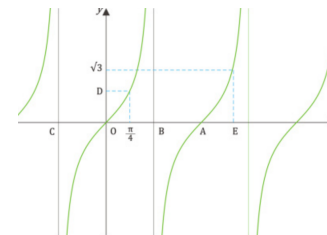
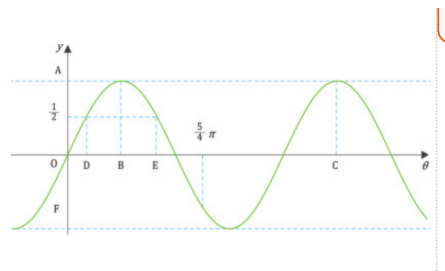
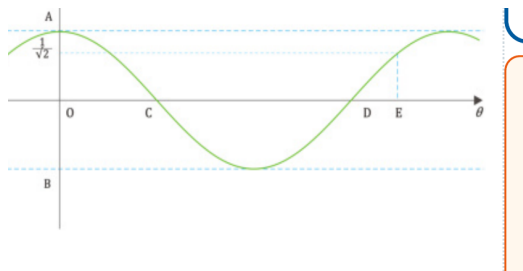


Classified Practice

Properties and Graphs of Trigonometric Functions

Q008 **Written**

Complete all four given graph tasks: draw the three parent graphs and find their periods; read the labelled scales on the given cosine graph; read the labelled scales on the given sine graph; and read the labelled scales on the given tangent graph.

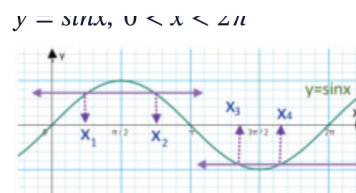


Classified Practice

Properties and Graphs of Trigonometric Functions

Q009 Written

The given graph represents $y = \sin x$ for $0 < x < 2\pi$. Find $x_1 + x_2 + x_3 + x_4$ from the four marked x-coordinates.



Then find: $x_1 + x_2 + x_3 + x_4$

Classified Practice

Properties and Graphs of Trigonometric Functions

Q010 Written

Draw the graphs of $y = 2 \cos \theta$ and $y = \tan \frac{\theta}{2}$, and find their periods.

Q011 Written

Draw the graphs of $y = 2 \sin \theta$, $y = \tan 2\theta$, and $y = \cos \frac{\theta}{2}$, and find their periods.

Classified Practice

Properties and Graphs of Trigonometric Functions

Q012 Written

Draw the nine given graphs and find each period: (a) $3 \sin \theta$, (b) $3 \cos \theta$, (c) $\frac{1}{2} \cos \theta$, (d) $\sin 3\theta$, (e) $\cos 2\theta$, (f) $\tan 3\theta$, (g) $\sin \frac{\theta}{2}$, (h) $\cos \frac{\theta}{3}$, (i) $\tan \frac{\theta}{3}$.

Classified Practice

Properties and Graphs of Trigonometric Functions

Q013 Written

Draw $y = -2|\sin(3x)|$ and find its period.

Q014 Written**Draw $y = \sin(\theta - \frac{\pi}{6})$ and find its period.**

Classified Practice

Properties and Graphs of Trigonometric Functions

Q015 Written

Draw the graphs and find their periods: (1) $y = \sin(\theta - \frac{\pi}{3})$, (2) $y = \cos(\theta + \frac{\pi}{4})$, (3) $y = \tan(\theta - \frac{\pi}{2})$.

Classified Practice

Properties and Graphs of Trigonometric Functions

Q016 Written

Draw all given graphs and find their periods: sine group $y = \sin(\theta + \frac{\pi}{4}), \sin(\theta - \frac{\pi}{2}), \sin(\theta + \frac{\pi}{3})$; **cosine group** $y = \cos(\theta - \frac{\pi}{3}), \cos(\theta - \frac{\pi}{4}), \cos(\theta + \frac{\pi}{6})$; **tangent group** $y = \tan(\theta + \frac{\pi}{6}), \tan(\theta - \frac{\pi}{3}), \tan(\theta - \frac{\pi}{4})$.

Classified Practice

Properties and Graphs of Trigonometric Functions

Q017 Written

Draw $y = 3 \cos \left| x + \frac{\pi}{3} \right|$ and find its period.

Q018 Written

Draw the graphs and find their periods: (1) $y = 2 \cos(2\theta - \frac{\pi}{2})$, (2) $y = \tan(\frac{\theta}{2} - \frac{\pi}{6})$.

Classified Practice

Properties and Graphs of Trigonometric Functions

Q019 Written

Draw the graphs and find their periods: (a) $y = 3 \sin \frac{\theta}{2}$, (b) $y = \tan(2\theta - \frac{\pi}{2})$, (c) $y = 2 \cos(2\theta - \frac{\pi}{3})$.

Classified Practice

Properties and Graphs of Trigonometric Functions

Q020 Written

Draw all nine given graphs and find their periods: (a) $2 \cos \frac{\theta}{2}$, (b) $3 \sin 2\theta$, (c) $2 \cos 2\theta$, (d) $2 \cos(\theta - \frac{\pi}{4})$, (e) $2 \sin(\theta - \frac{\pi}{3})$, (f) $\tan(2\theta - \frac{\pi}{3})$, (g) $3 \cos(2\theta - \frac{\pi}{2})$, (h) $3 \sin(2\theta - \frac{2\pi}{3})$, (i) $2 \cos(\frac{\theta}{2} - \frac{\pi}{4})$.

Classified Practice

Properties and Graphs of Trigonometric Functions

Q021 Written

Draw $y = 3 \sin |2x - \pi|$ and find its period.

Q022 MCQ $\sin(\theta + \pi)$ equals:**Choose the correct option.****A** $\sin \theta$ **B** $-\sin \theta$ **C** $\cos \theta$ **D** $-\cos \theta$

Classified Practice

Properties and Graphs of Trigonometric Functions

Q023 MCQ

 $\cos(\theta + \pi)$ equals:

Choose the correct option.

A $\cos \theta$ B $-\cos \theta$ C $\sin \theta$ D $-\sin \theta$

Q024 MCQ $\sin\left(\frac{\pi}{2} - \theta\right)$ equals:

Choose the correct option.

A $\sin \theta$ B $-\sin \theta$ C $\cos \theta$ D $-\cos \theta$

Classified Practice

Properties and Graphs of Trigonometric Functions

Q025 MCQ

 $\cos(-\theta)$ equals:

Choose the correct option.

A $\cos \theta$ B $-\cos \theta$ C $\sin \theta$ D $-\sin \theta$

Q026 MCQ **$\sin(-\theta)$ equals:****Choose the correct option.****A** $\sin \theta$ **B** $-\sin \theta$ **C** $\cos \theta$ **D** $-\cos \theta$

Classified Practice

Properties and Graphs of Trigonometric Functions

Q027 MCQ

 $\cos(\pi - \theta)$ equals:

Choose the correct option.

A $\cos \theta$ B $-\cos \theta$ C $\sin \theta$ D $-\sin \theta$

Q028 MCQ $\sin\left(\frac{3\pi}{2} + \theta\right)$ equals:

Choose the correct option.

A $\sin \theta$ B $-\sin \theta$ C $\cos \theta$ D $-\cos \theta$

Classified Practice

Properties and Graphs of Trigonometric Functions

Q029 MCQ

 $\tan(\theta - \frac{\pi}{2})$ equals:

Choose the correct option.

A $\tan \theta$ B $-\tan \theta$ C $\cot \theta$ D $-\cot \theta$

Q030 MCQ**The period of $y = \sin \theta$ is:****Choose the correct option.****A** $\pi/2$ **B** π **C** 2π **D** 4π

Classified Practice

Properties and Graphs of Trigonometric Functions

Q031 MCQ

The range of $y = \cos \theta$ is:**Choose the correct option.**

A $y \geq 0$

B $-1 \leq y \leq 1$

C $y \in \mathbb{R}$

D $y > 1$

Q032 MCQ

The period of $y = \tan \theta$ is:**Choose the correct option.**A $\pi/2$ B π C 2π D 4π

Classified Practice

Properties and Graphs of Trigonometric Functions

Q033 MCQ

A vertical asymptote of $y = \tan \theta$ occurs at:**Choose the correct option.**

A $\theta = k\pi$

B $\theta = \frac{\pi}{2} + k\pi$

C $\theta = 2k\pi$

D $\theta = \frac{\pi}{4} + k\pi$

Q034 MCQ

The maximum value of $\sin \theta$ is:**Choose the correct option.**

A -1

B 0

C 1

D π

Classified Practice

Properties and Graphs of Trigonometric Functions

Q035 MCQ

The x-coordinate of the first positive sine maximum is:**Choose the correct option.**

- A $\pi/4$
- B $\pi/2$
- C π
- D $3\pi/2$

Q036 MCQ**At $\theta = 0$, $\cos \theta$ equals:****Choose the correct option.****A** -1**B** 0**C** 1**D** undefined

Classified Practice

Properties and Graphs of Trigonometric Functions

Q037 MCQ

At $\theta = 0$, $\tan \theta$ equals:**Choose the correct option.**

- A -1
- B 0
- C 1
- D undefined

Q038 MCQ

The range of $y = 3 \sin \theta$ is:**Choose the correct option.**A $[-1,1]$ B $[-3,3]$ C $[0,3]$ D $\mathbb{R}$

Classified Practice

Properties and Graphs of Trigonometric Functions

Q039 MCQ

The period of $y = \sin 4\theta$ is:**Choose the correct option.**A $\pi/4$ B $\pi/2$ C π D 2π

Q040 MCQ**The period of $y = \cos(\theta/3)$ is:****Choose the correct option.**A $2\pi/3$ B 2π C 3π D 6π

Classified Practice

Properties and Graphs of Trigonometric Functions

Q041 MCQ

The period of $y = \tan 5\theta$ is:**Choose the correct option.**A $\pi/5$ B π C 5π D $2\pi/5$

Q042 MCQ

The range of $y = \frac{1}{2} \cos \theta$ is:

Choose the correct option.

A $[-2, 2]$

B $[-1, 1]$

C $[-1/2, 1/2]$

D $[0, 1/2]$

Classified Practice

Properties and Graphs of Trigonometric Functions

Q043 MCQ

The graph of $y = -2 \sin \theta$ is the parent sine graph:**Choose the correct option.**

- A shifted only
- B reflected and stretched vertically
- C compressed horizontally only
- D undefined

Q044 MCQ

The period of $y = \tan(\theta/2)$ is:**Choose the correct option.**A $\pi/2$ B π C 2π D 4π

Classified Practice

Properties and Graphs of Trigonometric Functions

Q045 MCQ

The period of $y = \sin(3\theta)$ is:**Choose the correct option.**A $\pi/3$ B $2\pi/3$ C 3π D 6π

Q046 MCQ**The graph of $y = \sin(\theta - \pi/4)$ is shifted:****Choose the correct option.**

- A left by $\pi/4$
 - B right by $\pi/4$
 - C up by $\pi/4$
 - D down by $\pi/4$
-
-
-

Classified Practice

Properties and Graphs of Trigonometric Functions

Q047 MCQ

The graph of $y = \cos(\theta + \pi/3)$ is shifted:**Choose the correct option.**A right by $\pi/3$ B left by $\pi/3$ C up by $\pi/3$ D down by $\pi/3$

Q048 MCQ**A horizontal shift changes the sine period from 2π to:****Choose the correct option.****A** π **B** 2π **C** 4π **D** 0

Classified Practice

Properties and Graphs of Trigonometric Functions

Q049 MCQ

A horizontal shift changes the tangent period from π to:**Choose the correct option.**A $\pi/2$ B π C 2π D 4π

Q050 MCQ**The graph of $y = \sin(\theta + \pi/2)$ equals:****Choose the correct option.**A $\sin \theta$ B $-\sin \theta$ C $\cos \theta$ D $-\cos \theta$

Classified Practice

Properties and Graphs of Trigonometric Functions

Q051 MCQ

The graph of $y = \cos(\theta - \pi)$ equals:**Choose the correct option.**A $\cos \theta$ B $-\cos \theta$ C $\sin \theta$ D $-\sin \theta$

Q052 MCQ

The graph of $y = \tan(\theta - \pi/2)$ has its central zero at:

Choose the correct option.

A $\theta = 0$

B $\theta = \pi/2$

C $\theta = \pi$

D $\theta = 2\pi$

Classified Practice

Properties and Graphs of Trigonometric Functions

Q053 MCQ

Since cosine is even, $\cos |x|$ equals:**Choose the correct option.**A $\cos(x)$ B $-\cos(x)$ C $\sin(x)$ D $|\cos(x)|$

Q054 MCQ**In $y = a \sin(b\theta - q)$, the horizontal shift is:****Choose the correct option.**

- A** q
- B** q/b
- C** b/q
- D** a/b

Classified Practice

Properties and Graphs of Trigonometric Functions

Q055 MCQ

The period of $y = 2 \cos(3\theta - \pi)$ is:**Choose the correct option.**A $\pi/3$ B $2\pi/3$ C π D 2π

Q056 MCQ

The range of $y = 4 \sin(2\theta - 1)$ is:**Choose the correct option.**A $[-1, 1]$ B $[-2, 2]$ C $[-4, 4]$ D $\mathbb{R}$

Classified Practice

Properties and Graphs of Trigonometric Functions

Q057 MCQ

For $y = \tan(2\theta - \pi/3)$, the shift is:**Choose the correct option.**

- A left by $\pi/3$
- B right by $\pi/6$
- C right by $\pi/3$
- D left by $\pi/6$

Q058 MCQ

The period of $y = 3 \sin(\theta/2)$ is:**Choose the correct option.**

- A π
 - B 2π
 - C 3π
 - D 4π
-
-
-

Classified Practice

Properties and Graphs of Trigonometric Functions

Q059 MCQ

The correct transformation order for $y = af(b\theta - q)$ starts with:

Choose the correct option.

- A vertical scale
- B horizontal shift
- C horizontal scale
- D reflection only

Q060 MCQ**The period of $y = \tan(4\theta + q)$ is:****Choose the correct option.****A** $\pi/4$ **B** $\pi/2$ **C** π **D** 2π

Classified Practice

Properties and Graphs of Trigonometric Functions

Q061 MCQ

The range of $y = 3 \sin |2x - \pi|$ is:**Choose the correct option.**A $[-3,3]$ B $[0,3]$ C $[-1,1]$ D $\mathbb{R}$

Q062 Written

Express $\cos \frac{11\pi}{6}$ as a function of an acute angle.

Classified Practice

Properties and Graphs of Trigonometric Functions

Q063 Written

Simplify $\sin(\theta + \pi) - \cos\left(\frac{\pi}{2} - \theta\right)$.

Q064 Written**Simplify** $\cos(\theta + \pi) + \cos(\frac{\pi}{2} - \theta)$.

Classified Practice

Properties and Graphs of Trigonometric Functions

Q065 Written

Evaluate $\tan\left(-\frac{3\pi}{4}\right)$ **using an acute reference angle.**

Q066 Written

State the period and range of $y = \cos \theta$, then list its key points on one cycle.

Classified Practice

Properties and Graphs of Trigonometric Functions

Q067 Written

State the period and asymptotes of $y = \tan \theta$ on $[-\pi, \pi]$.

Q068 Written

For the parent sine graph, find the x-values where $y = 0$ in $[0, 2\pi]$.

Classified Practice

Properties and Graphs of Trigonometric Functions

Q069 Written

For the parent cosine graph, find the x-values where $y = 0$ in $[0, 2\pi]$.

Q070 Written**Find the period and range of $y = 4 \cos \theta$.**

Classified Practice

Properties and Graphs of Trigonometric Functions

Q071 Written

Find the period and range of $y = \sin(5\theta)$.

Q072 Written**Find the period and range of $y = \tan(\theta/4)$.**

Classified Practice

Properties and Graphs of Trigonometric Functions

Q073 Written

Describe the transformations from $y = \cos \theta$ to $y = -\frac{1}{2} \cos(2\theta)$.

Q074 Written**Find the shift and period of $y = \sin(\theta + \frac{\pi}{6})$.**

Classified Practice

Properties and Graphs of Trigonometric Functions

Q075 Written

Find the shift and period of $y = \cos(\theta - \frac{\pi}{2})$.

Q076 Written**Find the central zero and period of $y = \tan(\theta + \frac{\pi}{4})$.**

Classified Practice

Properties and Graphs of Trigonometric Functions

Q077 Written

For $y = 2 \sin(4\theta - \pi)$, state the shift, period and range.

Q078 Written**For $y = \cos(\theta/3 + \pi/6)$, state the shift and period.**

Classified Practice

Properties and Graphs of Trigonometric Functions

Q079 Written

For $y = 3 \tan(2\theta + \pi)$, state the shift and period.

Q080 Written

For $y = 2 \cos(\theta/2 - \pi/8)$, state the shift, period and range.

Classified Practice

Properties and Graphs of Trigonometric Functions

Quiz MCQ 1 Concept Quiz · MCQ

The period of $y = \sin(2\theta - \frac{\pi}{3})$ is:**Choose the correct option.**A $\pi/2$ B π C 2π D 4π

Quiz MCQ 2 Concept Quiz · MCQ

$\sin(\theta + \pi)$ equals:

Choose the correct option.

A $\sin \theta$

B $-\sin \theta$

C $\cos \theta$

D $-\cos \theta$

Classified Practice

Properties and Graphs of Trigonometric Functions

Quiz MCQ 3 **Concept Quiz · MCQ****The range of $y = 2 \cos \theta$ is:****Choose the correct option.**A $[-1,1]$ B $[-2,2]$ C $[0,2]$ D $\mathbb{R}$

Quiz MCQ 4 Concept Quiz · MCQ

The graph of $y = \cos(\theta - \frac{\pi}{4})$ is shifted:

Choose the correct option.

- A left by $\frac{\pi}{4}$
- B right by $\frac{\pi}{4}$
- C up by $\frac{\pi}{4}$
- D down by $\frac{\pi}{4}$

Classified Practice

Properties and Graphs of Trigonometric Functions

Quiz WRITTEN 5 Concept Quiz · Written**For $y = 3 \sin(2\theta - \pi)$, state the horizontal shift, period and range.**

Quiz WRITTEN 6 Concept Quiz · Written

Simplify $\sin(\theta + \frac{\pi}{2}) + \sin(\theta + \pi)$.

Classified Practice

Properties and Graphs of Trigonometric Functions

Quiz WRITTEN 7 Concept Quiz · Written**Draw one cycle of $y = \tan(\theta - \frac{\pi}{3})$ and state its period and asymptotes.**

Quiz WRITTEN 8 Concept Quiz · Written

For $y = -2|\sin(3x)|$, state its period and range and describe the reflection.

Answer key

Use the question number shown on each page.

- Q001** (1) $-\sin \frac{\pi}{5}$ (2) 0
- Q002** (1) $\cos \frac{\pi}{6}$ (2) $-\sin \frac{\pi}{7}$ (3) $\sin^2 \theta - \cos^2 \theta$
- Q003** (1) $-\cos \frac{2\pi}{5}, -\cos \frac{2\pi}{7}, -\sin \frac{\pi}{8}, -\sin \frac{3\pi}{8}$ (2) $-\frac{1}{\sqrt{3}}, -1$ (3a) 0, (3b) $-\frac{1}{\sqrt{3}}$
- Q004** 0
- Q005** $\sin, \cos$: $T = 2\pi, -1 \leq y \leq 1$; $\tan$: $T = \pi, y \in \mathbb{R}$
- Q006** $A = 1, B = -1, C = \frac{\pi}{2}, D = 2\pi, E = \frac{1}{2}, F = \frac{5\pi}{4}$
- Q007** (1) $T_{\min} = 2\pi, T_{\max} = 2\pi, T_{\min} = \pi$; (2) read the marked extrema, intercepts and quarter-period points from the graph
- Q008** Use the standard cycles: sine/cosine have period 2π and range $[-1, 1]$; tangent has period π , range $\mathbb{R}$, and asymptotes at odd multiples of $\frac{\pi}{2}$. The requested A+F labels are read from the marked ticks and extrema in each supplied graph.
- Q009** Read x_1, x_2, x_3, x_4 from the supplied graph and add the four marked coordinates; the result is the graph-reading total.
- Q010** $y = 2 \cos \theta$: $T = 2\pi, -2 \leq y \leq 2$; $y = \tan \frac{\theta}{2}$: $T = 2\pi$
- Q011** $T = 2\pi, \frac{\pi}{2}, 4\pi$ respectively; ranges $[-2, 2], \mathbb{R}, [-1, 1]$
- Q012** (a) 2π (b) 2π (c) 2π (d) $\frac{2\pi}{3}$ (e) $\frac{\pi}{3}$ (f) $\frac{\pi}{3}$ (g) 4π (h) 6π (i) 3π ; ranges follow the outside coefficient
- Q013** $T = \frac{\pi}{3}, 0 \geq y \geq -2$
- Q014** shift right by $\frac{\pi}{6}, T = 2\pi$
- Q015** Shifts: right $\frac{\pi}{3}$, left $\frac{\pi}{4}$, right $\frac{\pi}{2}$; periods $2\pi, 2\pi, \pi$.
- Q016** All sine/cosine periods are 2π ; all tangent periods are π . The shifts are read directly from the brackets.
- Q017** $y = 3 \cos(x + \frac{\pi}{3})$ since cosine is even; $T = 2\pi$
- Q018** (1) $y = 2 \cos 2(\theta - \frac{\pi}{4}), T = \pi, -2 \leq y \leq 2$; (2) $y = \tan \frac{1}{2}(\theta - \frac{\pi}{3}), T = 2\pi$
- Q019** (a) $T = 4\pi, -3 \leq y \leq 3$; (b) $T = \frac{\pi}{2}$; (c) $T = \pi, -2 \leq y \leq 2$
- Q020** Periods: (a) 4π ; (b) π ; (c) π ; (d) 2π ; (e) 2π ; (f) $\frac{\pi}{2}$; (g) π ; (h) π ; (i) 4π . Ranges are set by the outside factors.
- Q021** $y = 3|\sin(2x - \pi)|, T = \frac{\pi}{2}, 0 \leq y \leq 3$
- Q022** B
- Q023** B
- Q024** C
- Q025** A
- Q026** B
- Q027** B
- Q028** D
- Q029** D
- Q030** C
- Q031** B
- Q032** B
- Q033** B
- Q034** C
- Q035** B
- Q036** C
- Q037** B
- Q038** B
- Q039** B
- Q040** D
- Q041** A
- Q042** C
- Q043** B
- Q044** C
- Q045** B
- Q046** B
- Q047** B
- Q048** B
- Q049** B
- Q050** C
- Q051** B
- Q052** B
- Q053** A
- Q054** B
- Q055** B
- Q056** C
- Q057** B
- Q058** D
- Q059** B
- Q060** A
- Q061** B
- Q062** $\cos \frac{\pi}{6}$
- Q063** $-\sin \theta - \sin \theta = -2 \sin \theta$
- Q064** $-\cos \theta + \sin \theta$
- Q065** 1
- Q066** $T = 2\pi, -1 \leq y \leq 1$; (0, 1), $(\frac{\pi}{2}, 0), (\pi, -1), (\frac{3\pi}{2}, 0), (2\pi, 1)$
- Q067** $T = \pi; \theta = \pm \frac{\pi}{2}$
- Q068** 0, $\pi, 2\pi$
- Q069** $\frac{\pi}{2}, \frac{3\pi}{2}$
- Q070** $T = 2\pi, -4 \leq y \leq 4$
- Q071** $T = \frac{2\pi}{5}, -1 \leq y \leq 1$
- Q072** $T = 4\pi, y \in \mathbb{R}$
- Q073** horizontal compression by 1/2, reflection in x-axis and vertical scale 1/2; $T = \pi$
- Q074** left by $\frac{\pi}{6}, T = 2\pi$
- Q075** right by $\frac{\pi}{2}, T = 2\pi$
- Q076** $\theta = -\frac{\pi}{4}; T = \pi$
- Q077** right by $\frac{\pi}{4}, T = \frac{\pi}{2}, -2 \leq y \leq 2$
- Q078** left by $\frac{\pi}{2}, T = 6\pi$
- Q079** left by $\frac{\pi}{2}, T = \frac{\pi}{2}$
- Q080** right by $\frac{\pi}{4}, T = 4\pi, -2 \leq y \leq 2$
- Quiz MCQ 1** B
- Quiz MCQ 2** B
- Quiz MCQ 3** B

Quiz MCQ 4 B**Quiz WRITTEN 6** $\cos \theta - \sin \theta$ **Quiz WRITTEN 7** $T = \pi; \theta = \frac{5\pi}{6} + k\pi$ **Quiz WRITTEN 8** $T = \frac{\pi}{3}, -2 \leq y \leq 0$; reflected below the**Quiz WRITTEN 5** right by $\frac{\pi}{2}$, $T = \pi, -3 \leq y \leq 3$

Concept 4 | Solving Equations and Inequalities Involving Trigonometric Functions

English Student Edition • Focused formulas and guided practice

Formula opener

Sine

$$\sin x = \sin \alpha$$

Cosine

$$\cos x = \cos \alpha$$

Tangent

$$\tan x = \tan \alpha$$

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Q001 Written

Solve (a) $-2 \cos \theta = \sqrt{2}$ and (b) $\tan \theta + \sqrt{3} = 0$, giving the general solution for (b).

Q002 Written

Solve (a) $2 \cos \theta - 1 = 0$ and (b) $\tan \theta - 1 = 0$ for $0 \leq \theta \leq 2\pi$.

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Q003 Written

Solve all given equations on $0 \leq \theta \leq 2\pi$: (a) $2 \sin \theta + 1 = 0$, (b) $2 \sin \theta = \sqrt{3}$, (c) $3 \sin \theta + 3 = 0$, (d) $\sqrt{2} \cos \theta + 1 = 0$, (e) $2 \cos \theta = \sqrt{3}$, (f) $2 \cos \theta - \sqrt{2} = 0$, (g) $\tan \theta + 1 = 0$, (h) $\sqrt{3} \tan \theta = 1$, (i) $\sqrt{3} \tan \theta = 3$, **and give general solutions for** (j) $2 \sin \theta + \sqrt{3} = 0$, (k) $\cos \theta + 1 = 0$, (l) $\sqrt{3} \tan \theta + 3 = 0$.

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Q004 **Written**

Solve $\tan^2 \theta - (2 + \sqrt{3}) \tan \theta + 2\sqrt{3} = 0$ and give the general solution.

Q005 Written

Solve for $0 \leq \theta < 2\pi$: (a) $2 \cos \theta > -\sqrt{3}$, (b) $\tan \theta \leq \frac{1}{\sqrt{3}}$.

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Q006 Written

Solve for $0 \leq \theta < 2\pi$: (a) $2 \sin \theta \leq -1$, (b) $\cos \theta > \frac{1}{2}$, (c) $\tan \theta > 1$.

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Q007 Written

Solve all given inequalities for $0 \leq \theta < 2\pi$: (a) $\sin \theta > \frac{\sqrt{3}}{2}$, (b) $\sin \theta - \frac{\sqrt{2}}{2} > 0$, (c) $2 \sin \theta + \sqrt{3} \leq 0$, (d) $\cos \theta > 0$, (e) $\cos \theta - \frac{1}{2} \leq 0$, (f) $2 \cos \theta + \sqrt{2} \leq 0$, (g) $\tan \theta \leq -\frac{1}{\sqrt{3}}$, (h) $\tan \theta + \sqrt{3} > 0$, (i) $\tan \theta + 1 \geq 0$.

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Q008 Written

Solve $2 \sin^2 \theta - \cos \theta - 1 > 0$ **for** $0 \leq \theta < 2\pi$.

Q009 Written**Solve for $0 \leq \theta < 2\pi$: (a) $2 \cos^2 \theta - 3 \sin \theta = 0$, (b) $2 \sin^2 \theta - 3 \cos \theta - 3 \leq 0$.**

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Q010 Written

Solve for $0 \leq \theta < 2\pi$: (a) $2 \sin^2 \theta - \sin \theta = 0$, (b) $2 \cos^2 \theta - \sin \theta = 2$, (c) $2 \sin^2 \theta - 4 < 5 \cos \theta$.

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Q011 Written

Solve all given equations and inequalities for $0 \leq \theta < 2\pi$: (a) $2 \sin^2 \theta + \sin \theta - 1 = 0$, (b) $2 \sin^2 \theta + 3 \cos \theta = 0$, (c) $2 \sin^2 \theta - \cos \theta = 2$, (d) $2 \cos^2 \theta - 5 \cos \theta - 3 = 0$, (e) $2 \cos^2 \theta + \sin \theta - 1 = 0$, (f) $2 \cos^2 \theta + 3 \sin \theta = 3$, (g) $2 \cos^2 \theta + 3 \sin \theta \leq 0$, (h) $2 \sin^2 \theta + \cos \theta > 1$, (i) $2 \cos^2 \theta - 4 \leq 5 \sin \theta$.

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Q012 Written

Solve $\frac{1-2\sin^2\theta+\sin\theta}{\cos^2\theta} \geq 1$ for $0 \leq \theta < 2\pi$.

Q013 Written

Solve for $0 \leq \theta < 2\pi$: (a) $\sin(\theta + \frac{\pi}{3}) = \frac{1}{2}$, (b) $\sin(\theta + \frac{\pi}{6}) > -\frac{1}{2}$.

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Q014 Written

Solve for $0 \leq \theta < 2\pi$: (a) $\sin(\theta + \frac{\pi}{6}) = \frac{1}{2}$, (b) $\tan(\theta + \frac{\pi}{4}) = \sqrt{3}$, (c) $\cos(\theta + \frac{\pi}{4}) < -\frac{\sqrt{2}}{2}$.

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Q015 Written

Solve for $0 \leq \theta < 2\pi$: (a) $\cos(\theta + \frac{\pi}{3}) = -\frac{\sqrt{3}}{2}$, (b) $\cos(\theta + \frac{\pi}{6}) = -\frac{1}{2}$, (c) $\sin(\theta + \frac{\pi}{4}) = \frac{\sqrt{3}}{2}$, (d) $\tan(\theta + \frac{\pi}{4}) = -1$, (e) $\tan(\theta + \frac{\pi}{3}) = \frac{1}{\sqrt{3}}$, (f) $\tan(\theta + \frac{\pi}{6}) = 1$, (g) $\sin(\theta + \frac{\pi}{6}) \leq \frac{\sqrt{3}}{2}$, (h) $\sin(\theta + \frac{\pi}{4}) \geq \frac{\sqrt{2}}{2}$, (i) $\cos(\theta + \frac{\pi}{3}) > \frac{1}{2}$.

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Q016 Written

Solve $4 \sin^2(\theta + \frac{\pi}{6}) \geq 1$ for $0 \leq \theta < 2\pi$.

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Q017 Written

A student divides $\tan x \times \cos x = \frac{1}{2}$ by $\cos x$, obtaining $\tan x = \frac{1}{2 \cos x}$. Evaluate this strategy and propose a better substitution on $[0^\circ, 360^\circ[$.

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Q018 MCQ

The period used for a general tangent solution is:**Choose the correct option.**A 2π B π C $\pi/2$ D 4π

Q019 MCQ**The solutions of $2 \cos \theta = 1$ on $[0, 2\pi]$ are:****Choose the correct option.****A $\pi/6, 11\pi/6$** **B $\pi/3, 5\pi/3$** **C $2\pi/3, 4\pi/3$** **D $\pi/4, 7\pi/4$**

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Q020 MCQ

Which root is impossible when solving $2 \sin^2 \theta + 3 \sin \theta - 2 = 0$?

Choose the correct option.

A $\sin \theta = 1/2$

B $\sin \theta = -2$

C $\sin \theta = 0$

D $\sin \theta = -1/2$

Q021 MCQ

The quadratic challenge factors into roots:**Choose the correct option.**

- A 1 and $\sqrt{3}$
- B 2 and $\sqrt{3}$
- C 2 and 3
- D $\pi/3$ and 2

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Q022 MCQ

For $2 \cos \theta > -\sqrt{3}$, the excluded arc is:

Choose the correct option.

A $[0, 5\pi/6]$

B $[5\pi/6, 7\pi/6]$

C $[7\pi/6, 2\pi)$

D $(\pi/6, 5\pi/6)$

Q023 MCQ**The inequality $\tan \theta > 1$ on $[0, 2\pi)$ holds on:****Choose the correct option.****A** $(\pi/4, \pi/2) \cup (5\pi/4, 3\pi/2)$ **B** $[0, \pi/4]$ **C** $(\pi/2, \pi)$ **D** $[\pi, 2\pi)$

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Q024 MCQ

A closed endpoint in a trig inequality is used when the sign is:

Choose the correct option.

- A strictly greater
- B strictly less
- C greater than or equal to
- D never

Q025 MCQ**The 5-14 challenge reduces to which cosine condition?****Choose the correct option.**

A $\cos \theta > 1/2$

B $-1 < \cos \theta < 1/2$

C $\cos \theta \leq -1$

D $\cos \theta \geq 1$

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Q026 MCQ

When a quadratic gives $\sin \theta = -2$, we:**Choose the correct option.**

A keep it

B reject it

C square it

D divide by 2

Q027 MCQ

The first step for $2 \cos^2 \theta - 3 \sin \theta = 0$ is to use:**Choose the correct option.**

A $\sin^2 \theta + \cos^2 \theta = 1$

B $\tan \theta = \sin \theta$

C $\cos \theta = 1$

D the tangent period

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Q028 MCQ

The solution of $2 \sin^2 \theta + \sin \theta - 1 = 0$ includes:

Choose the correct option.

A $\theta = \pi/6$

B $\theta = \pi/4$

C $\theta = \pi/3$

D $\theta = 2\pi/3$

Q029 MCQ**In the rational challenge, which angle must be excluded?****Choose the correct option.****A** 0**B** $\pi/4$ **C** $\pi/2$ **D** π

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Q030 MCQ

After setting $t = \theta + \pi/3$ with $0 \leq \theta < 2\pi$, the t-range is:

Choose the correct option.

A $[0, 2\pi)$

B $[\pi/3, 7\pi/3)$

C $[-\pi/3, 5\pi/3)$

D $[\pi, 3\pi)$

Q031 MCQ**The solutions of $\tan(\theta + \pi/4) = \sqrt{3}$ include:****Choose the correct option.**A $\pi/12$ B $\pi/6$ C $\pi/3$ D $\pi/2$

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Q032 MCQ

For $\sin(\theta + \pi/4) \geq \sqrt{2}/2$, the θ -solution is:

Choose the correct option.

A $[0, \pi/2]$

B $(0, \pi/2)$

C $[\pi/2, \pi]$

D $[0, 2\pi)$

Q033 MCQ**The 5-16 challenge is equivalent to:****Choose the correct option.**

A $|\sin(t)| \geq 1/2$

B $\sin(t) = 0$

C $\cos(t) > 1/2$

D $\tan(t) = 1$

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Q034 MCQ

In the Reflect problem, $\tan x \cos x$ simplifies to:**Choose the correct option.**A $\cos(x)$ B $\sin(x)$ C $\tan^2 x$

D 1

Quiz MCQ 1 **Concept Quiz · MCQ**

The general solution of $\tan \theta = -1$ is:

Choose the correct option.

A $-\pi/4 + 2n\pi$

B $-\pi/4 + n\pi$

C $\pi/4 + 2n\pi$

D $\pi/2 + n\pi$

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Quiz MCQ 2 **Concept Quiz · MCQ**

The solution of $\cos \theta > 0$ on $[0, 2\pi)$ is:

Choose the correct option.

A $(\pi/2, 3\pi/2)$

B $[0, \pi/2) \cup (3\pi/2, 2\pi)$

C $[0, \pi)$

D $(\pi, 2\pi)$

Quiz MCQ 3 Concept Quiz · MCQ

The valid sine root of $2 \sin^2 \theta + 3 \sin \theta - 2 = 0$ is:

Choose the correct option.

A $\sin \theta = -2$

B $\sin \theta = 1/2$

C $\sin \theta = 2$

D $\sin \theta = -1/2$

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Quiz MCQ 4 [Concept Quiz · MCQ](#)

The first operation in a shifted-argument problem is to:

Choose the correct option.

- A divide by $\cos \theta$
- B set t equal to the whole argument and find its range
- C change radians to degrees
- D square both sides

Quiz WRITTEN 5 Concept Quiz · Written**Solve** $2 \cos \theta = \sqrt{3}$ for $0 \leq \theta < 2\pi$.

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Quiz WRITTEN 6 Concept Quiz · Written**Solve** $\tan \theta \leq -1$ for $0 \leq \theta < 2\pi$.

Quiz WRITTEN 7 Concept Quiz · Written

Solve $2 \sin^2 \theta + \sin \theta - 1 = 0$ **for** $0 \leq \theta < 2\pi$.

Classified Practice

Solving Equations and Inequalities Involving Trigonometric Functions

Quiz WRITTEN 8 Concept Quiz · Written**Solve** $\cos(\theta + \frac{\pi}{3}) > \frac{1}{2}$ **for** $0 \leq \theta < 2\pi$.

Answer key

Use the question number shown on each page.

Q001 (a) $\theta = \frac{3\pi}{4}, \frac{5\pi}{4}$ (b) $\theta = \frac{2\pi}{3} + n\pi, n \in \mathbb{Z}$ **Q012** $0 \leq \theta < \frac{\pi}{2}$ or $\frac{\pi}{2} < \theta \leq \pi$ **Q020** B

Q002 (a) $\theta = \frac{\pi}{3}, \frac{5\pi}{3}$ (b) $\theta = \frac{\pi}{4}, \frac{5\pi}{4}$ **Q013** (a) $\theta = \frac{\pi}{2}, \frac{11\pi}{6}$ (b) $0 \leq \theta < \pi$ or $\frac{5\pi}{3} < \theta < 2\pi$ **Q021** B

Q003 (a) $\frac{3\pi}{4}, \frac{11\pi}{4}$ (b) $\frac{\pi}{2}, \frac{3\pi}{2}$ (c) $\frac{3\pi}{4}, \frac{5\pi}{4}$ (d) $\frac{3\pi}{4}, \frac{5\pi}{4}$ (e) $\frac{\pi}{2}, \frac{3\pi}{2}$ (f) $\frac{\pi}{2}, \frac{3\pi}{2}$ (g) $\frac{\pi}{2}, \frac{3\pi}{2}$ (h) $\frac{\pi}{2}, \frac{3\pi}{2}$ (i) $\frac{\pi}{2}, \frac{3\pi}{2}$ (j) $\frac{\pi}{2}, \frac{3\pi}{2}$ (k) $\frac{\pi}{2}, \frac{3\pi}{2}$ (l) $\frac{\pi}{2}, \frac{3\pi}{2}$ (m) $\frac{\pi}{2}, \frac{3\pi}{2}$ (n) $\frac{\pi}{2}, \frac{3\pi}{2}$ (o) $\frac{\pi}{2}, \frac{3\pi}{2}$ (p) $\frac{\pi}{2}, \frac{3\pi}{2}$ (q) $\frac{\pi}{2}, \frac{3\pi}{2}$ (r) $\frac{\pi}{2}, \frac{3\pi}{2}$ (s) $\frac{\pi}{2}, \frac{3\pi}{2}$ (t) $\frac{\pi}{2}, \frac{3\pi}{2}$ (u) $\frac{\pi}{2}, \frac{3\pi}{2}$ (v) $\frac{\pi}{2}, \frac{3\pi}{2}$ (w) $\frac{\pi}{2}, \frac{3\pi}{2}$ (x) $\frac{\pi}{2}, \frac{3\pi}{2}$ (y) $\frac{\pi}{2}, \frac{3\pi}{2}$ (z) $\frac{\pi}{2}, \frac{3\pi}{2}$ **Q014** (a) $\theta = 0, \frac{2\pi}{3}$ (b) $\theta = \pi, \frac{13\pi}{12}$ (c) $\frac{\pi}{2} < \theta < \pi$ **Q022** B

Q004 $\tan \theta = 2$ or $\tan \theta = \sqrt{3}$; $\theta = \arctan(2) + n\pi$ or $\theta = \frac{\pi}{3} + n\pi$ **Q015** (a) $\frac{\pi}{2}, \frac{5\pi}{6}$ (b) $\frac{\pi}{2}, \frac{7\pi}{6}$ (c) $\pi, \frac{5\pi}{12}$ (d) $\frac{\pi}{2}, \frac{3\pi}{2}$ (e) $\frac{\pi}{2}, \frac{3\pi}{2}$ (f) $\frac{\pi}{2}, \frac{3\pi}{2}$ (g) $\frac{\pi}{2}, \frac{3\pi}{2}$ (h) $\frac{\pi}{2}, \frac{3\pi}{2}$ (i) $\frac{\pi}{2}, \frac{3\pi}{2}$ (j) $\frac{\pi}{2}, \frac{3\pi}{2}$ (k) $\frac{\pi}{2}, \frac{3\pi}{2}$ (l) $\frac{\pi}{2}, \frac{3\pi}{2}$ (m) $\frac{\pi}{2}, \frac{3\pi}{2}$ (n) $\frac{\pi}{2}, \frac{3\pi}{2}$ (o) $\frac{\pi}{2}, \frac{3\pi}{2}$ (p) $\frac{\pi}{2}, \frac{3\pi}{2}$ (q) $\frac{\pi}{2}, \frac{3\pi}{2}$ (r) $\frac{\pi}{2}, \frac{3\pi}{2}$ (s) $\frac{\pi}{2}, \frac{3\pi}{2}$ (t) $\frac{\pi}{2}, \frac{3\pi}{2}$ (u) $\frac{\pi}{2}, \frac{3\pi}{2}$ (v) $\frac{\pi}{2}, \frac{3\pi}{2}$ (w) $\frac{\pi}{2}, \frac{3\pi}{2}$ (x) $\frac{\pi}{2}, \frac{3\pi}{2}$ (y) $\frac{\pi}{2}, \frac{3\pi}{2}$ (z) $\frac{\pi}{2}, \frac{3\pi}{2}$ **Q023** A

Q005 (a) $0 \leq \theta < \frac{5\pi}{6}$ or $\frac{7\pi}{6} < \theta < 2\pi$ (b) $0 \leq \theta \leq \frac{\pi}{6}$ or $\frac{\pi}{2} < \theta \leq \frac{7\pi}{6}$ or $\frac{3\pi}{2} < \theta < 2\pi$ **Q016** $0 \leq \theta \leq \frac{2\pi}{3}$ or $\pi \leq \theta \leq \frac{5\pi}{3}$ **Q024** C

Q006 (a) $\frac{7\pi}{6} \leq \theta \leq \frac{11\pi}{6}$ (b) $0 \leq \theta < \frac{\pi}{3}$ or $\frac{5\pi}{3} < \theta < 2\pi$ (c) $\frac{\pi}{4} < \theta < \frac{\pi}{2}$ or $\frac{5\pi}{4} < \theta < \frac{3\pi}{2}$ **Q017** Dividing by $\cos(x)$ separates the interacting factors and is not helpful, **Q025** B

Q007 (a) $\frac{\pi}{2}, \frac{3\pi}{2}$ (b) $\frac{\pi}{2}, \frac{3\pi}{2}$ (c) $\frac{\pi}{2}, \frac{3\pi}{2}$ (d) $\frac{\pi}{2}, \frac{3\pi}{2}$ (e) $\frac{\pi}{2}, \frac{3\pi}{2}$ (f) $\frac{\pi}{2}, \frac{3\pi}{2}$ (g) $\frac{\pi}{2}, \frac{3\pi}{2}$ (h) $\frac{\pi}{2}, \frac{3\pi}{2}$ (i) $\frac{\pi}{2}, \frac{3\pi}{2}$ (j) $\frac{\pi}{2}, \frac{3\pi}{2}$ (k) $\frac{\pi}{2}, \frac{3\pi}{2}$ (l) $\frac{\pi}{2}, \frac{3\pi}{2}$ (m) $\frac{\pi}{2}, \frac{3\pi}{2}$ (n) $\frac{\pi}{2}, \frac{3\pi}{2}$ (o) $\frac{\pi}{2}, \frac{3\pi}{2}$ (p) $\frac{\pi}{2}, \frac{3\pi}{2}$ (q) $\frac{\pi}{2}, \frac{3\pi}{2}$ (r) $\frac{\pi}{2}, \frac{3\pi}{2}$ (s) $\frac{\pi}{2}, \frac{3\pi}{2}$ (t) $\frac{\pi}{2}, \frac{3\pi}{2}$ (u) $\frac{\pi}{2}, \frac{3\pi}{2}$ (v) $\frac{\pi}{2}, \frac{3\pi}{2}$ (w) $\frac{\pi}{2}, \frac{3\pi}{2}$ (x) $\frac{\pi}{2}, \frac{3\pi}{2}$ (y) $\frac{\pi}{2}, \frac{3\pi}{2}$ (z) $\frac{\pi}{2}, \frac{3\pi}{2}$ **Q026** B

Q008 $\frac{\pi}{3} < \theta < \pi$ or $\pi < \theta < \frac{5\pi}{3}$ use $\tan(x) \cos(x) = \sin(x)$ (with **Q027** A

Q009 (a) $\theta = \frac{\pi}{6}, \frac{5\pi}{6}$ (b) $0 \leq \theta \leq \frac{2\pi}{3}$ or $\theta = \pi$ or $\frac{4\pi}{3} \leq \theta < 2\pi$ $\cos(x) \neq 0$, so $\sin(x) = \frac{1}{2}$ and **Q028** A

Q010 (a) $\theta = 0, \frac{\pi}{6}, \frac{5\pi}{6}, \pi$ (b) $\theta = 0, \pi, \frac{7\pi}{6}, \frac{11\pi}{6}$ (c) $0 \leq \theta < \frac{2\pi}{3}$ or $\frac{4\pi}{3} < \theta < 2\pi$ $x = 30^\circ, 150^\circ$. **Q029** C

Q011 (a) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (b) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (c) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (d) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (e) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (f) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (g) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (h) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (i) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (j) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (k) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (l) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (m) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (n) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (o) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (p) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (q) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (r) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (s) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (t) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (u) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (v) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (w) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (x) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (y) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ (z) $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$ **Q030** B

Q018 B **Q019** B **Q031** A

Q032 A

Q033 A

Q034 B

Quiz MCQ 1 B

Quiz MCQ 2 B

Quiz MCQ 3 B

Quiz MCQ 4 B

Quiz WRITTEN 5 $\theta = \frac{\pi}{6}, \frac{11\pi}{6}$

Quiz WRITTEN 6 $\frac{\pi}{2} < \theta \leq \frac{3\pi}{4}$ or $\frac{3\pi}{2} < \theta \leq \frac{7\pi}{4}$

Quiz WRITTEN 7 $\theta = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2}$

Quiz WRITTEN 8 $\frac{4\pi}{3} < \theta < 2\pi$

Concept 5 | Advanced Theorems and Applications

English Student Edition • Focused formulas and guided practice

Formula opener

Addition

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$

Double angle

$$\sin 2A = 2 \sin A \cos A$$

Cosine

$$\cos 2A = \cos^2 A - \sin^2 A$$

Classified Practice

Advanced Theorems and Applications

Q001 Written

Find the maximum and minimum of $y = \sin^2\theta + \cos\theta + 1$, $0 \leq \theta < 2\pi$, and the corresponding θ .

Q002 Written**Find the maximum and minimum of $y = \cos^2\theta - \sin\theta$, $0 \leq \theta < 2\pi$, and the corresponding θ .**

Classified Practice

Advanced Theorems and Applications

Q003 Written

Solve the four given maximum/minimum problems on $0 \leq \theta < 2\pi$, including $y = \sin^2 \theta + \sin \theta - 1$, $y = \cos^2 \theta - \cos \theta + 1$, $y = 2 \sin \theta + \cos^2 \theta - 1$, and $y = -2 \sin^2 \theta - 2 \cos \theta + 1$. Include the given challenge $y = \cos^2 \theta - \sin \theta$.

Classified Practice

Advanced Theorems and Applications

Q004 **Written****Use addition theorems to find $\sin 75^\circ$ and $\cos(\pi/12)$.**

Q005 Written

Given $\sin \alpha = \frac{\sqrt{3}}{2}$ **in QII** and $\cos \beta = -\frac{3}{5}$ **in QIII**, **find** $\sin(\alpha + \beta)$ **and** $\cos(\alpha - \beta)$.

Classified Practice

Advanced Theorems and Applications

Q006 Written

Complete the given addition-theorem exercises *exact values, quadrant – controlled α, β , and the three – angle challenge* ($\sin \alpha = 0.6, \cos \beta = 0.8, \tan \gamma = 0.75$).

Q007 Written

Find $\tan 105^\circ$, then find the angle between $y = 2x + 1$ and $y = -3x + 2$.

Classified Practice

Advanced Theorems and Applications

Q008 **Written**

Find the angle between each given pair of straight lines, including $y = \frac{1}{3}x$ with $y = 2x$, and $8x + 2y - 3 = 0$ with $-5x + 3y + 1 = 0$.

Classified Practice

Advanced Theorems and Applications

Q009 Written

Find the exact angles for all given line pairs and solve the challenge: a line through $P(2, 3)$ makes acute angle $\tan \theta = 3/4$ with $x + 2y - 4 = 0$; give both line equations and the areas of the triangles with the y -axis.

Classified Practice

Advanced Theorems and Applications

Q010 Written

Given $\sin \alpha = 1/3$ with α in QII, find $\sin 2\alpha$, $\cos 2\alpha$, $\tan 2\alpha$.

Q011 Written

Given $\sin \alpha = -2/3$ **with** $2\pi < \alpha < 3\pi$, **find** $\sin 2\alpha$, $\cos 2\alpha$, $\tan 2\alpha$.

Classified Practice

Advanced Theorems and Applications

Q012 **Written**

Solve the given double-angle exercises for given sine/cosine values and the challenge $\cos \theta = 0.6, \pi < \theta < 2\pi$, evaluating $\cos 3\theta \tan 2\theta$.

Q013 Written

Given $\cos \alpha = 3/5$, $0 < \alpha < \pi$, **find** $\sin(\alpha/2)$, $\cos(\alpha/2)$, $\tan(\alpha/2)$, **and evaluate** $\cos(5\pi/8)$.

Classified Practice

Advanced Theorems and Applications

Q014 **Written**

Given $\cos \alpha = -1/3$, $0 < \alpha < \pi$, **find** $\sin(\alpha/2)$, $\cos(\alpha/2)$, $\tan(\alpha/2)$, **and evaluate** $\sin(3\pi/8)$.

Q015 Written

Solve the given half-angle exercises for the three quadrant intervals and the challenge with $\tan A = \frac{\sqrt{5}}{2}, 4\pi < A < 5\pi,$
 $\cos B = -\frac{\sqrt{5}}{3}, 5\pi < B < 6\pi.$

Classified Practice

Advanced Theorems and Applications

Q016 Written

Solve on $0 \leq \theta < 2\pi$: $a \sin 2\theta + \sin \theta = 0$, $b \cos 2\theta - 7 \cos \theta + 4 \geq 0$.

Q017 Written**Solve the given equation** $\sin 2\theta + \cos \theta = 0$ **on** $0 \leq \theta < 2\pi$.

Classified Practice

Advanced Theorems and Applications

Q018 **Written**

Solve all given double-angle equations and inequalities on $0 \leq \theta < 2\pi$, including the challenge $\cos 2\theta - 7 \cos \theta + 4 \geq 0$.

Q019 Written

Rewrite $\sin \theta - \sqrt{3} \cos \theta$ as $r \sin(\theta + \alpha)$, $r > 0$, $-\pi < \alpha < \pi$.

Classified Practice

Advanced Theorems and Applications

Q020 Written

Rewrite the **given** expressions $\sin \theta - \cos \theta$, $\sin \theta + \cos \theta$, $3 \sin \theta - \sqrt{3} \cos \theta$, and $3 \sin \theta - \cos \theta$ in **harmonic-addition form**.

Q021 Written

Rewrite all given harmonic-addition exercises in the form $r \sin(\theta + \alpha)$, including $2 \sin \theta + 2 \cos \theta$ and the challenge $\sin \theta - \sqrt{3} \cos \theta$.

Classified Practice

Advanced Theorems and Applications

Q022 Written

Solve on $0 \leq \theta < 2\pi$: $a \sin \theta - \cos \theta = 1$, $b \sin \theta + \cos \theta < 3/2$.

Q023 Written**Solve** $\sin \theta \leq \cos \theta$ **on** $0 \leq \theta < 2\pi$.

Classified Practice

Advanced Theorems and Applications

Q024 Written

Solve all given harmonic-addition equations and inequalities on $0 \leq \theta < 2\pi$, including $\sin \theta + \cos \theta + 1 = 0$ and $\sin \theta - \cos \theta + 3/2 > 0$.

Q025 Written

Find the maximum and minimum of $y = 2 \sin(\theta - \pi/3)$, and of $y = 2 \cos 2\theta + 4 \cos \theta + 1$, $0 \leq \theta < 2\pi$.

Classified Practice

Advanced Theorems and Applications

Q026 Written

Find the maximum and minimum of the given functions $y = -\frac{4}{3} \sin^2 \theta + \cos \theta$ and $y = \cos 2\theta - 2 \cos \theta$.

Q027 Written

Solve all given maximum/minimum problems in the final lesson, including absolute-value and double-angle forms, and state the corresponding angles where requested.

Classified Practice

Advanced Theorems and Applications

Q028 MCQ

The substitution for $\sin^2 \theta + \cos \theta + 1$ should be:

Choose the correct option.

A $t = \sin \theta$

B $t = \cos \theta$

C $t = \tan \theta$

D $t = \theta$

Q029 MCQ**The maximum of $\sin^2 \theta + \cos \theta + 1$ is:****Choose the correct option.****A** 0**B** 1**C** 9/4**D** 4

Classified Practice

Advanced Theorems and Applications

Q030 MCQ

An extremal vertex is valid only if t lies in:**Choose the correct option.**A $[-2,2]$ B $[-1,1]$ C $[0,1]$

D all real numbers

Q031 MCQ

The identity for $\sin(a+b)$ is:**Choose the correct option.**

- A $\sin \alpha \cos \beta + \cos \alpha \sin \beta$
- B $\sin \alpha \sin \beta$
- C $\cos \alpha \cos \beta$
- D $\tan \alpha + \tan \beta$

Classified Practice

Advanced Theorems and Applications

Q032 MCQ

 $\sin 75$ degrees equals:**Choose the correct option.**A $(\sqrt{6}-\sqrt{2})/4$ B $(\sqrt{6}+\sqrt{2})/4$ C $1/2$ D $\sqrt{3}/2$

Q033 MCQ**A missing trig value is selected using:****Choose the correct option.**

- A the quadrant
- B the denominator
- C the period only
- D a graph scale

Classified Practice

Advanced Theorems and Applications

Q034 MCQ

The slope of $y=mx+k$ is:**Choose the correct option.**

A k

B m

C $1/m$ D $m+k$

Q035 MCQ**The angle formula uses denominator:****Choose the correct option.****A** $1+m_1m_2$ **B** m_1-m_2 **C** m_1+m_2 **D** m_1m_2

Classified Practice

Advanced Theorems and Applications

Q036 MCQ

For an acute angle, if $\tan \theta_1$ is negative we use:

Choose the correct option.

A θ_1

B $\pi - \theta_1$

C $2\theta_1$

D $-\theta_1$ only

Q037 MCQ **$\sin(2a)$ equals:****Choose the correct option.**

- A $\sin \alpha$
- B $2 \sin \alpha \cos \alpha$
- C $\cos^2 \alpha$
- D $\tan \alpha$

Classified Practice

Advanced Theorems and Applications

Q038 MCQ

 $\cos(2\alpha)$ can be written as:**Choose the correct option.**A $\cos^2 \alpha - \sin^2 \alpha$ B $2\sin \alpha$ C $\tan \alpha$ D $1 + \sin \alpha$

Q039 MCQ**For theta in QIV, sin theta is:****Choose the correct option.**

- A positive
- B negative
- C zero always
- D undefined

Classified Practice

Advanced Theorems and Applications

Q040 MCQ

The half-angle formula for $\sin(a/2)$ uses:**Choose the correct option.**

- A $(1 - \cos \alpha)/2$
- B $(1 + \cos \alpha)/2$
- C $\tan \alpha$
- D $\cos 2\alpha$

Q041 MCQ**The sign of a half-angle root is chosen from:****Choose the correct option.**

- A the original quadrant interval
- B the coefficient only
- C the denominator
- D the period of $\tan$

Classified Practice

Advanced Theorems and Applications

Q042 MCQ

If $\cos a = -1/3$ and $0 < a < \pi$, $\cos(a/2)$ is:

Choose the correct option.

A negative

B positive

C zero

D undefined

Q043 MCQ**For double-angle equations, first rewrite in:****Choose the correct option.**

- A one trig function
- B two unrelated angles
- C degrees only
- D a logarithm

Classified Practice

Advanced Theorems and Applications

Q044 MCQ

The factorisation of $\sin(2\theta) + \sin\theta$ is:**Choose the correct option.**A $\sin\theta(2\cos\theta + 1)$ B $\cos\theta(2\sin\theta - 1)$ C $\sin\theta + \cos\theta$ D $2\sin\theta$

Q045 MCQ**A tangent asymptote is always:****Choose the correct option.**

- A included
- B excluded
- C a maximum
- D an endpoint

Classified Practice

Advanced Theorems and Applications

Q046 MCQ

For $a \sin \theta + b \cos \theta$, r is:**Choose the correct option.**A $a+b$ B $\sqrt{a^2+b^2}$ C $a-b$ D ab

Q047 MCQ**The harmonic target form is:****Choose the correct option.****A** $r \sin(\theta + \alpha)$ **B** $r \cos \theta$ **C** $a \tan \theta$ **D** $\sin \theta$ only

Classified Practice

Advanced Theorems and Applications

Q048 MCQ

The angle a is located from point:**Choose the correct option.**

- A $P(a,b)$
 - B $P(r,\theta)$
 - C the origin only
 - D the x intercept
-
-
-

Q049 MCQ**After synthesis, t is usually:****Choose the correct option.****A** $\theta + \alpha$ **B** θ^2 **C** $\tan \theta$ **D** $\cos \theta$

Classified Practice

Advanced Theorems and Applications

Q050 MCQ

If $\max(\sin \theta + \cos \theta) = \sqrt{2}$, then $\sin \theta + \cos \theta < \frac{3}{2}$ is:

Choose the correct option.

- A never true
- B always true
- C true only at π
- D undefined

Q051 MCQ**In a strict inequality, equality endpoints are:****Choose the correct option.**

- A included
- B excluded
- C doubled
- D random

Classified Practice

Advanced Theorems and Applications

Q052 MCQ

The range of $2\sin(\theta - \pi/3)$ is:**Choose the correct option.**A $[-1,1]$ B $[-2,2]$ C $[0,2]$ D $[-3,3]$

Q053 MCQ**For a double-angle maximum problem, use:****Choose the correct option.**

- A the double-angle formula then a bounded variable
- B a logarithm
- C only a graph
- D no substitution

Classified Practice

Advanced Theorems and Applications

Q054 MCQ

A vertex outside $[-1,1]$ is:**Choose the correct option.**

- A an allowed extremum
- B rejected
- C always the maximum
- D always the minimum

Quiz MCQ 1 Concept Quiz · MCQ

The maximum of $2 \sin(\theta - \pi/3)$ is:

Choose the correct option.

A -2

B -1

C 1

D 2

Classified Practice

Advanced Theorems and Applications

Quiz MCQ 2 Concept Quiz · MCQ

The addition-theorem value $\sin 75^\circ$ is:

Choose the correct option.

A $(\sqrt{6} + \sqrt{2})/4$

B $(\sqrt{6} - \sqrt{2})/4$

C $1/2$

D 1

Quiz MCQ 3 Concept Quiz · MCQ

For $y = 2x^2 + 4x + 1$, the vertex x-value is:

Choose the correct option.

A -2

B -1

C 1

D 2

Classified Practice

Advanced Theorems and Applications

Quiz MCQ 4 **Concept Quiz · MCQ**

The angle between slopes 2 and -3 is:

Choose the correct option.

A $\pi/6$

B $\pi/4$

C $\pi/3$

D $\pi/2$

Quiz WRITTEN 5 Concept Quiz · Written

Find the maximum and minimum of $y = \sin^2 \theta + \cos \theta + 1$ on $0 \leq \theta < 2\pi$.

Classified Practice

Advanced Theorems and Applications

Quiz WRITTEN 6 Concept Quiz · Written

Find $\sin 2\alpha$, $\cos 2\alpha$, $\tan 2\alpha$ when $\sin \alpha = 1/3$ and α is in QII.

Quiz WRITTEN 7 Concept Quiz · Written

Rewrite $3 \sin \theta - \sqrt{3} \cos \theta$ **as** $r \sin(\theta + \alpha)$.

Classified Practice

Advanced Theorems and Applications

Quiz WRITTEN 8 Concept Quiz · Written**Solve** $\sin \theta \leq \cos \theta$ **on** $0 \leq \theta < 2\pi$.

Answer key

Use the question number shown on each page.

- Q001** $\max y = \frac{9}{4}$ at $\theta = \frac{\pi}{3}, \frac{5\pi}{3}$; $\min y = 0$ at $\theta = \frac{2\pi}{3}, \frac{4\pi}{3}$
- Q002** $\max y = \frac{5}{4}$ at $\theta = \frac{7\pi}{6}, \frac{11\pi}{6}$; $\min y = -1$ at $\theta = \frac{\pi}{2}, \frac{3\pi}{2}$
- Q003** See the numbered solution: substitute $t = \sin \theta$ or $t = \cos \theta$, complete the square, then test $t = \pm 1$.
- Q004** $\sin 75^\circ = \cos \pi/12 = \frac{\sqrt{6} + \sqrt{2}}{4}$
- Q005** $\sin \alpha + \beta = \frac{4 - 3\sqrt{3}}{10}$; $\cos \alpha - \beta = \frac{3 - 4\sqrt{3}}{10}$
- Q006** See the numbered solution; the three-angle challenge gives $\sin(\alpha + \beta + \gamma) = \frac{117}{125}$.
- Q007** $\tan 105^\circ = -2 + \sqrt{3}$; $\theta = \frac{\pi}{4}$
- Q008** The first pair gives $\theta = \frac{\pi}{4}$; the second pair also gives $\theta = \frac{\pi}{4}$.
- Q009** See the numbered solution; the two slopes are $m = 2/11, -2$, with triangle areas $49/165$ and $25/3$.
- Q010** $\sin 2\alpha = -\frac{4\sqrt{2}}{9}$, $\cos 2\alpha = \frac{7}{9}$, $\tan 2\alpha = -\frac{4\sqrt{2}}{7}$
- Q011** $\sin 2\alpha = \frac{4\sqrt{5}}{9}$, $\cos 2\alpha = \frac{1}{9}$, $\tan 2\alpha = 4\sqrt{5}$
- Q012** For the challenge, $\cos 3\theta = -\frac{117}{125}$, $\tan 2\theta = \frac{24}{7}$, so the product is $-\frac{2808}{875}$.
- Q013** $\sin \frac{\alpha}{2} = \frac{1}{\sqrt{5}}$, $\cos \frac{\alpha}{2} = \frac{2}{\sqrt{5}}$, $\tan \frac{\alpha}{2} = \frac{1}{2}$, $\cos \frac{5\pi}{8} = -\frac{\sqrt{2-\sqrt{2}}}{2}$
- Q014** $\sin \frac{\alpha}{2} = \sqrt{\frac{2}{3}}$, $\cos \frac{\alpha}{2} = \frac{1}{\sqrt{3}}$, $\tan \frac{\alpha}{2} = \sqrt{2}$, $\sin \frac{3\pi}{8} = \frac{\sqrt{2+\sqrt{2}}}{2}$
- Q015** Use the signed half-angle formulas; for the challenge, $\cos(A/2 - B/2) = \frac{\sqrt{3+\sqrt{5}} - \sqrt{15-5\sqrt{5}}}{6}$, $\sin(A/4) = \sqrt{\frac{1-\sqrt{5/6}}{2}}$.
- Q016** $a\theta = 0, \frac{2\pi}{3}, \pi, \frac{4\pi}{3}$; $b\frac{\pi}{3} \leq \theta \leq \frac{5\pi}{3}$
- Q017** $\theta = \frac{\pi}{2}, \frac{7\pi}{6}, \frac{3\pi}{2}, \frac{11\pi}{6}$
- Q018** See the numbered solution: replace double angles, solve in $s = \sin \theta$ or $c = \cos \theta$, and translate the accepted ranges.
- Q019** $r = 2$, $\alpha = -\frac{\pi}{3}$, $\sin \theta - \sqrt{3} \cos \theta = 2 \sin \theta - \frac{\pi}{3}$
- Q020** $\sqrt{2} \sin \theta - \frac{\pi}{4}$, $\sqrt{2} \sin \theta + \frac{\pi}{4}$, $2\sqrt{3} \sin \theta - \frac{\pi}{6}$, $\sqrt{10} \sin \theta - \arctan \frac{1}{2}$
- Q021** See the numbered solution; every result has $r = \sqrt{a^2 + b^2} > 0$ and α chosen in $(-\pi, \pi)$.
- Q022** $a\theta = \frac{\pi}{2}, \pi$; $b\ 0 \leq \theta < 2\pi$
- Q023** $0 \leq \theta \leq \frac{\pi}{4}$ or $\frac{5\pi}{4} \leq \theta < 2\pi$
- Q024** See the numbered solution; synthesize first, then solve in the shifted variable over its full range.
- Q040** A
- Q041** A
- Q042** B
- Q043** A
- Q044** A
- Q045** B
- Q046** B
- Q047** A
- Q048** A
- Q049** A
- Q050** B
- Q051** B
- Q052** B
- Q053** A
- Q054** B
- Quiz MCQ 1** D
- Quiz MCQ 2** A
- Quiz MCQ 3** A
- Quiz MCQ 4** B
- Quiz WRITTEN 5** $\max = \frac{9}{4}, \min = 0$
- Quiz WRITTEN 6** $-\frac{4\sqrt{2}}{9}, \frac{7}{9}, -\frac{4\sqrt{2}}{7}$
- Quiz WRITTEN 7** $2\sqrt{3} \sin \theta - \frac{\pi}{6}$
- Quiz WRITTEN 8** $[0, \frac{\pi}{4}] \cup [\frac{5\pi}{4}, 2\pi)$